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HYPERSONIC JET IMPINGEMENT DURING LANDING ON LUNAR AND MARTIAN SURFACES

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Declaration of Originality and Submission Information

I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements.

Abstract

This study aims to investigate the flow dynamics and performance of two supersonic nozzles and propose a mitigation strategy for plume-surface interactions in the context of Martian descent technologies. A comprehensive literature review was conducted to establish a solid foundation of knowledge and understanding of the current state-of-the-art in this field. Two nozzles were successfully simulated using computational fluid dynamics (CFD) techniques. The first nozzle, operating at Mach 4.84, was analyzed to gain insights into its flow characteristics and performance. The second nozzle, the Martian Descent Thruster Nozzle, was simulated at Mach 7.525 and validated against theoretical predictions. The computational approach demonstrated remarkable accuracy in capturing the complex flow dynamics and key features, such as shock structures and expansion waves, in both nozzles.

Furthermore, a novel mitigation strategy for plume-surface interactions was proposed, focusing on the introduction of microjets to reduce the impingement pressure and alter the flow field. The groundwork for this strategy was laid through a comprehensive methodology that involved leveraging data from initial nozzle simulations and utilizing an iterative process to determine the optimal parameters for the microjets. Despite the promising groundwork established for the microjet mitigation strategy, time constraints prevented the complete acquisition of simulation results. However, the foundational work completed in this study, including the development of computational tools, data collection techniques, and the identification of key parameters, provides a solid platform for future research and optimization of this approach. The outcomes of this study contribute to the advancement of supersonic nozzle design and plume-surface interaction mitigation strategies. The validated computational models and insights gained from the Mach 4.84 and Mach 7.525 nozzles can inform future design decisions and performance optimization efforts. Additionally, the groundwork laid for the microjet mitigation strategy opens up new avenues for research and development in the field of plume control and spacecraft propulsion.

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Acronyms

CAD Computer Aided Design. 20, 21

CFD Computational Fluid Dynamics. 5, 19, 21

EVA Extravehicular Activity. 9

ISRU In-Situ Resource Utilization. 1

LM Lander Module. 3, 5, 6, 9

MOC Method of Characteristics. 15, 25, 26

NASA National Aeronautics and Space Administration. 1, 10

NS Navier-Stokes. 19

PAL (Ames Research Center) Planetary Aeolian Laboratory. 8

PM Prandtl-Meyer. 15

PSI Plume Surface Interactions. 1–11, 13, 14, 37

TRL Technology Readiness Level. 3, 4, 13

Nomenclature

β	Thermal Expansion Coefficient
ϵ	Nozzle Expansion Ratio
Γ	Specific Heat Ratio
λ	Thermal Conductivity
ϕ	Viscous Dissipation Factor
ρ	Density
τ	Viscous Shear Stress Tensor
A^*	Nozzle Throat Area
A_e	Nozzle Exit Area
A_e/A^*	Nozzle Expansion Ratio
d	(Primary) Nozzle Exit Diameter
d_{inj}	Injector Nozzle Exit Diameter
h/d	Normalised Altitude
M_e	Nozzle Exit Temperature
P_a	Ambient Pressure
P_c	(Primary) Thruster Chamber Pressure
P_e	Nozzle Exit Pressure
P_g	Ground Pressure due to Impingement
T_c	Thruster Chamber Temperature
T_e	Nozzle Exit Temperature
x	Axial Length along Plume or Nozzle Axis

1 Introduction

> "I think dust is probably one of our greatest inhibitors to a nominal operation on the Moon. I think we can overcome other physiological or physical or mechanical problems except dust."
> — Gene Cernan, Apollo 17 Technical Debrief

1.1 From Space Exploration to Planetary Surface Challenges

Space exploration epitomises the pinnacle of human ingenuity and curiosity, symbolising our relentless pursuit to fathom and master the vast enigmas of the universe. This journey of discovery and technological advancement has evolved significantly from the nascent stages of space travel. The Soviet Union's launch of *Sputnik 1*, the inaugural artificial satellite, inaugurated the era of extraterrestrial exploration (Kuznetsov et al. 2015). This milestone was succeeded by the historic achievement of Russian Lieutenant Yuri Gagarin, who became the first human to orbit Earth aboard *Vostok 1*. Nevertheless, the *Apollo 11* mission stands as a watershed moment in the annals of space exploration. On July 20, 1969, at 20:17 UTC, astronauts Neil A. Armstrong, Michael Collins, and Edwin E. Aldrin Jr. accomplished an unparalleled feat by being the first humans to set foot on the Moon (*Apollo 11 Postflight Analysis and Mission Report* 1970). The intense rivalry between the USSR and the USA during this epoch, set against the backdrop of the Cold War, underscored the space race's importance in asserting technological and intellectual supremacy (*The Soviet Space Program* 1969). This spirit of competition continues to influence the contemporary landscape of space exploration. Recent advancements have witnessed an expansion in the number of nations successfully reaching the Moon, with the USSR and the USA now joined by Japan, China, and notably, India (AS 2024). India's mission to the lunar south pole, particularly, signifies a pivotal milestone in global space exploration. The presence of water ice at the lunar poles is crucial, as it supports life, enables thermal regulation, and facilitates In-Situ Resource Utilization (ISRU) fuel production, thereby enhancing the sustainability of future missions (Xu et al. 2003). Importantly, this ice also serves as a vital scientific resource, offering profound insights into the Moon's geological evolution and cosmic heritage (National Aeronautics and Space Administration 2019).

The contemporary resurgence in space exploration, particularly concerning *lunar* and *Martian* missions, has been substantially invigorated by technological advancements, strategic geopolitical aspirations, and an expanding consortium of international and commercial entities endeavoring to capitalise on the scientific and economic potentialities of space (Bergan 2022). Eminent initiatives such as National Aeronautics and Space Administration (NASA)'s Artemis program, which aspires to re-establish human presence on the Moon and ensure a sustainable foothold by the decade's conclusion, are poised to serve as a proving ground for prospective manned expeditions to Mars. Analogously, the Chinese Lunar Exploration Program and the European Space Agency's Lunar Pathfinder mission epitomise the global commitment to the exploration and utilisation of *extraterrestrial* environments. Notwithstanding substantial investments by preeminent spacefaring nations such as the United States and China (Statista 2023), these audacious extraterrestrial ventures are beset with numerous technological impediments. Paramount among these is the ubiquitous issue of Plume Surface Interactions (PSI), which is a universally shared conundrum across all space agencies, representing a particularly intractable challenge (Ketsdever et al. 2005; Roberts and Hastings 2015; Gimelshein and Levin 2010). This dilemma, central to the discourse of space exploration, constitutes a pivotal juncture, necessitating further disucssion in the ensuing discussion.

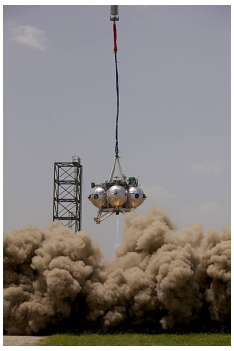
The investigation of PSI, or *Plume Surface Interactions*, constitutes a pivotal component of contemporary space exploration, focusing on the complex dynamics engendered by high-enthalpy gas plumes emitted by spacecraft retro-propulsion engines during the critical terminal phase of landing. These plumes, traversing at supersonic or hypersonic velocities, impact planetary surfaces, engendering a multifaceted interplay of gas dynamics and soil physics. As an inherently multiphase and multisystem discipline, PSI delineates the environmental conditions surrounding landers due to the impingement of high-temperature rocket exhaust

on the regolith of celestial bodies such as the Moon and Mars (Mantovani et al. 2022). The intricate interactions within plume flow physics, erosion physics, and ejecta dynamics give rise to the formation of a robust recirculation region that fluidises the surface and ejects loose granular material. The ramifications of these interactions are manifold, presenting significant risks including the destabilisation of the lander module, potential damage to exposed hardware from high-velocity debris, diminished visibility, and the possible spoofing of landing sensors. Multiple shock-shock and shock-surface interactions within the PSI milieu result in the lofting and propulsion of regolith, altering surface topography and inducing deeper excavation of subsurface soils. This dynamic process not only modifies the landscape but also accelerates regolith particles to velocities that can imperil the structural integrity of landing modules and pre-existing surface installations. Given the intensified mission planning to the Moon and Mars, comprehending and mitigating the effects of PSI are paramount to the success of future extraterrestrial missions. Consequently, the study of PSI has ascended to a focal point of aerospace research, driven by the significant challenges these *phenomena* pose during spacecraft landings on such celestial bodies. By addressing the intricate gas dynamics and soil physics involved, researchers aim to devise strategies to ameliorate the adverse effects of PSI, thereby ensuring safer and more successful landings on celestial bodies, such as the Moon and Mars. The work of Manish Mehta (2019) underscores the criticality of this research, emphasising the necessity for a comprehensive understanding of PSI to safeguard landers and surface installations from the potentially severe consequences of these interactions.

Figure 1 elucidates the distinct interactions of regolith across various planetary surfaces, accentuating how the phenomena of PSI diverge markedly between planets due to disparities in atmospheric conditions and soil properties, among other factors. These variations are detailed by Philip Metzger, J. Lane, et al. (2021) and are particularly emphasised by M. Mehta, Sengupta, et al. (2013) within the Martian context. Panel (a) of Figure 1 illustrates an initiative by *Masten Space Systems* to expedite the creation of lunar landing sites using rocket engine exhaust and aluminium oxide particles. The image conspicuously reveals a pronounced proclivity for dust dispersion in the *vacuum* conditions of the Moon's surface, critical observations that Zook (1975) and A. E. Metzger et al. (1973), along with others, elaborate upon later in *Section 2.1.1*. Conversely, Figure 1 (b) captures the *1.5 'Bravo' Morpheus* during a translational Tether Test conducted in August 2013. The ensuing visualisation from this test indicates that PSI on Mars tends to engender more concentrated and extensive dust clouds. This phenomenon is thoroughly documented by M. Mehta, Sengupta, et al. (2013). The intricate nature of these interactions is further expounded by International Agency Working Group (2016), who contend that solutions efficacious on one planetary surface may not necessarily be applicable to another, thereby accentuating the multifaceted understanding requisite for addressing dust mitigation across diverse celestial bodies.



(a) Lunar Surface Testing.



(b) Martian Surface Testing.

Figure 1: Visualisation of Regolith Dispersal on Lunar and Martian Surfaces. Provided by Planetary Data System (2022).

Mantovani et al. (2022) elucidated that PSI encompasses three pivotal sub-disciplines impacting spacecraft during landing: *Plume* Physics, *Erosion* Physics, and *Ejecta* Dynamics. Within the field of Ejecta Dynamics, prior discourses have highlighted the potential hazards posed by airborne debris during lunar landings, such as obscured visibility, compromised structural integrity, and the potential for navigational systems to malfunction. Figure 2 below delineates the various trajectories of such debris as a Lander Module (LM) descends on the lunar surface, necessitating further investigation into their impact on the wider lunar environment. Erosion Physics examines the effects of soil and debris that destabilise landers at touchdown, potentially damaging the structure and violating operational constraints. Plume Physics addresses the interactions of exhaust gases with the planetary surface, which can precipitate aerodynamic destabilisation and thermal challenges during descent. These sub-disciplines are essential for developing numerical models that predict and mitigate adverse effects on space missions.

The primary emphasis of this study, however, will be on Plume Physics, which is pivotal in understanding the interactions between the spacecraft's exhaust plume and the planetary surface. Plume Physics encompasses a panoply of phenomena, such as the impingement of the exhaust gases on the surface, the entrainment of loose soil particles, and the formation of shock waves consequent to the high-velocity exhaust flow (M. Mehta, Sengupta, et al. 2013). Of particular pertinence to this study are the initial phenomenon, the impingement of exhaust gases, and the terminal phenomenon, the formation of shock waves, both of which are integral to comprehending the intricate dynamics of PSI. These interactions can lead to significant challenges during the landing phase, including loss of visibility, increased thermal loads on the spacecraft, and potential instability caused by the altered flow field around the lander (Manish Mehta 2019).

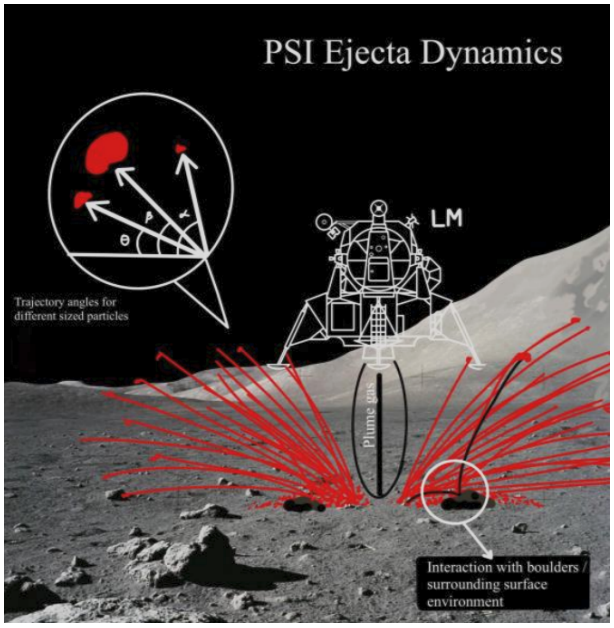


Figure 2: Illustration of PSI Ejecta Dynamics on Lunar Surface. Provided by Bueno (2022).

Mitigation strategies for regolith on *lunar* and *Martian* surfaces are indispensable, albeit beset with formidable developmental challenges. Afshar-Mohajer et al. (2015) highlighted an array of techniques, including mechanical brushes, advanced surfaces, gas and liquid solutions, and electrostatic methods, which have been proposed to address this exigent issue (J. R. Gaier 2005; Kawamoto and Inoue 2008; Calle et al. 2011; Margiotta et al. 2012). Nevertheless, many of these solutions necessitate further testing and refinement to achieve higher Technology Readiness Level (TRL)'s. Considering the distinct environmental conditions on the Moon and Mars, as explored by International Agency Working Group (2016), a technology efficacious on one celestial body may not be applicable on the other, thereby requiring adaptable solutions

tailored to the idiosyncratic conditions of each planetary surface. Moreover, effective mitigation strategies often necessitate a multifaceted approach to address the plethora of challenges posed by planetary dust. Predominantly, current methodologies focus on the removal of accumulated dust rather than precluding its deposition, which can precipitate operational impediments and damage to equipment (P. T. Metzger, J. E. Lane, et al. 2016). Additionally, many proposed strategies, such as constructing artificial landing surfaces with erosion-resistant materials and erecting boundary walls to shield equipment from plume flow, entail significant engineering challenges and logistical constraints. The transportation and implementation of these technologies on the Moon or Mars remain a considerable hurdle. In summation, while several promising strategies exist, there is an unequivocal need for more rigorous testing and development to enhance their effectiveness and adaptability. Addressing these challenges is paramount to ensuring the success of future missions and the safeguarding of vital equipment and infrastructure on both the Moon and Mars. Table 12 is presented in **Appendix** A for clarity on different classifications of TRL.

1.2 Project Aims and Objectives

The principal aim of this study is to develop and validate an intricate numerical approach for the simulation of Hypersonic Jet Impingement During Landing on Lunar and Martian Surfaces. By achieving this objective, we shall not only enhance the credibility of our subsequent results but also gain crucial insights that will serve as a catalyst for the further refinement of our sophisticated methods. By successfully completing this endeavor, we aim to deepen our understanding of the intricate interactions, fundamental physics, and core principles that underpin the studies of PSI and high-speed flow. Through this enhanced comprehension, we seek to unravel the complexities inherent in these domains which will pave way for the subsequent analysis.

Thus, building upon these foundations, the secondary aim of this study is to investigate the integration of microjets as a means of mitigating Plume Surface Interactions (PSI) during spacecraft landings on lunar and Martian surfaces. In theory, microjets can introduce controlled perturbations into the boundary and shear layers of the primary jet, enhancing mixing and energy redistribution. This integration is expected to destabilise the boundary layer, promoting early transition to turbulence, which could reduce peak heat fluxes and shear stresses on impinged surfaces. Additionally, microjets are anticipated to modify shear layers, leading to a more uniform spreading of the jet and diffusing kinetic energy over a larger area, thereby reducing localized impact forces and heating. The generation of vortical structures by microjets may enhance ambient gas entrainment, mixing, and diffusion rates, resulting in a more distributed pressure profile that mitigates concentrated impact pressures. Furthermore, by injecting microjets at angles to the primary flow, it is theorized that the primary jet can be deflected away from critical surfaces, creating a more favorable pressure distribution on the landing surface. Optimising the activation conditions of supersonic microjets is also expected to ensure they can be turned on and off as needed, achieving optimal flow control without degrading the spacecraft's operational performance. This comprehensive investigation into the fluid dynamics of microjet implementation aims to advance the development of effective PSI mitigation strategies for space exploration missions.

2 Literature Review

2.1 Contemporary Knowledge of PSI Phenomenology

This literature review endeavors to furnish an exhaustive overview of PSI phenomena, commencing with documented effects and advancing toward an in-depth comprehension of the underlying physical principles. The operational challenges engendered by PSI will be examined, succeeded by a discourse on contemporary mitigation strategies. Ultimately, a novel approach to mitigating PSI will be proposed, predicated upon the insights derived from the preceding sections.

2.1.1 Observed Plume-Surface Effects on the Moon

The intricate interplay between spacecraft exhaust plumes and the lunar surface during descent manoeuvres commands scholarly attention due to its complex impact dynamics, a subject of considerable academic investigation. As expounded by Zook (1975), these interactions catalyse the formation of *"secondary ejecta"*, a term denoting the lunar soil particles displaced by the forces exerted by the descending spacecraft. The propulsion of these particles is influenced not only by the spacecraft's size and descent velocity but also by the Moon's unique environmental conditions. In the absence of a substantial lunar atmosphere, exhaust plumes experience unrestrained expansion, significantly diverging from terrestrial conditions and thus producing distinct surface interactions characterised by the displacement of *regolith*. The regolith itself, subjected to exhaustive characterisation in the studies by R. V. Morris (1985) and A. E. Metzger et al. (1973), exhibits a heterogeneous morphology. Predominantly composed of particulate matter with the majority of grains measuring less than one centimeter in diameter, the regolith also includes a variety of larger formations such as angular blocks and glass shards. Additionally, it encompasses unique textures, described as "ropey or Swiss-cheese," which are interspersed with submicron voids and fissures. This morphological diversity substantially increases the regolith's reactive surface area, thereby influencing its response to the kinetic interactions with exhaust plumes.

Philip Metzger, J. Lane, et al. (2021) discerned that the propulsion of particulate matter, ranging from dust to sand-sized granules, transpires at relatively elevated velocities, exerting significant alterations on the lunar surface. Immer et al. (2011) elucidated this phenomenon further, reporting that *ultra-fine* lunar soil particles are capable of being ejected at velocities exceeding the lunar escape velocity ($\approx 2.4\ km/s$). This assertion is substantiated by NASA (2020), who corroborated these observations from the Apollo missions, wherein ejecta velocities were measured at an exit velocity of 3.1 km/s. Such velocities facilitated extensive dispersal across the lunar surface and posited the potential for some particles to enter heliocentric orbits. Empirical investigations by Immer et al. (2011) into the *Apollo 12* mission detailed the plume's propensity for ejecting particles at low trajectory angles, estimated within the approximate range of 1° to 3°. These findings align with subsequent observations from other Apollo missions, conducted by the same author, as depicted in Figure 31 in **Appendix** D, where an average angle of 3.3° is documented across five distinct *Apollo* missions. Findings by You et al. (2021) from the Chang'E-4 mission, which recorded higher dust ejection angles-up to 7°-as the LM neared the surface, suggests a broader distribution of ejected materials. It is plausible to infer that these interactions, particularly at lower angles, will consequently propel the regolith at increased velocities.

Philip Metzger and J. Lane (2012) carried out a study, with Computational Fluid Dynamics (CFD) amongst other techniques utilised, which, interestingly validated Immer et al. (2011) findings. Ejecta was observed to travel at $\approx 1°$ to 4° for most stages of descent. However, as You et al. (2021) noted earlier, particles were also observed to comprise larger angle trajectories, precisely angles of $\approx 15°$, particularily at distances closer to the surface. Key findings indicated that the velocities of ejected particles during lunar landings vary based on several factors: particle size, the altitude of the lander above the lunar surface, the lander's thrust (which itself depends on the lander's mass), and the initial distance of the particles from the lander at the moment of lift-off. Highlighting that smaller particle sizes would be expected to be affected by

PSI at higher descent altitudes, as seen in Figure 3 below, where particle sizes of 70 μm and 3.5mm were observed to be ejected when the LM was positioned vertically at distances of 20m and 15m from the lunar surface, respectively.

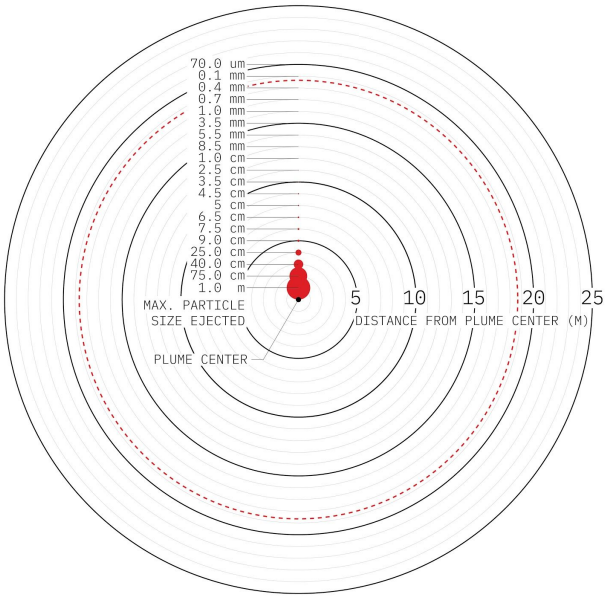


Figure 3: Polar chart showing particle sizes propelled by exhaust at 40ft intervals from the epicenter. The dotted boundary indicates the maximum range (926 microns) beyond which particles settle out of the airstream. Provided by Susant and P. Metzger (2016).

Watkins et al. (2021) elucidates that the hypersonic flow of the plume begins to exert an influence on the lunar surface when spacecraft attain altitudes approximately between 30 and 40 meters. This critical juncture is underscored by You et al. (2021), who identifies this moment as pivotal, marking the point at which surface shearing forces surpass the static friction of the lunar soil, thereby initiating significant erosional processes. Notably, the observations of astronauts during the *Apollo 12* and *Apollo 14* missions, as documented by Scott et al. (1970), corroborate Watkins' assertions, with dust clouds being observed at altitudes ranging from 32 to 40 meters above the surface. As the Lander Module continues its descent, concentrating the engine's blast onto a progressively smaller area, dust particles achieve greater radial speeds and distances. Detailed investigations, such as those by A. B. Morris et al. (2015), employed a hybrid Navier–Stokes/Direct Simulation Monte Carlo (DSMC) solver for the gas flow and a DSMC solver for the solid phase flow to simulate axisymmetric plume impingement and dust dispersion on the lunar surface. These simulations revealed that particles measuring 800 nm in size could be propelled to velocities approaching 1000 m/s, a relatively high speed theorised by Zook (1975) and subsequently validated by Immer et al. (2011). Furthermore, during the *Apollo 14* mission, Figure 4 overleaf, depicts a post-landing photograph illustrating the extent of surface erosion directly beneath the engine bell. Mitchell et al. (1971) described how each Apollo landing generated a distinct erosion track, culminating in a shallow pit where the Lander Module hovered momentarily, displacing approximately 666 kg of lunar soil and demonstrating the physical impact of the landing thrust. These landings also influence the lunar regolith's structure. As McKay et al. (1986) notes, the repeated impact of plumes alters the regolith's composition, as evidenced by the formation of *breccias*, fragmented rock material, resulting from shock over-pressures and heat, thus adding to the regolith's complexity.

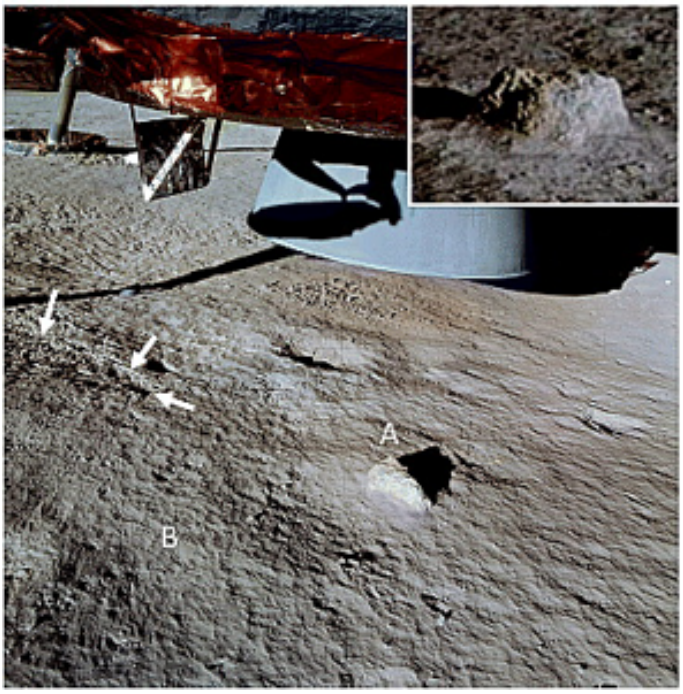


Figure 4: Arrows show the boundary where the upper soil layer was uniformly removed. Rock A, beneath the nozzle, is visible. Surface B has a hummocky texture from plume interaction. Inset confirms Rock A's partial embedment. Provided by Philip Metzger, Smith, et al. (2011).

Based on his, previously discussed, findings, P. T. Metzger (2020) later derived a model based on available empirical data which predicted quantities of each particle size, their velocities, and impact angles for each location on the Moon, scaled by lander thrust trajectory curve and distance to landing site. You et al. (2021) provided further analysis on footage from China's, 1.2 ton, Chang'e 4 lander, indicating that 19kg of material was ejected during the descent phase. These findings validated P. T. Metzger (2020) model showing the amount of ejecta scaled as vehicle mass to the 2.5 power. Naturally, a discerning reader would intuitively conclude that the effects of PSI, particularly on lunar surfaces, extend beyond the immediate vicinity of the lander to influence distant locations as well. In the same study, Immer et al. (2011) discovered that particle ejection during the *Apollo 12* landing led to the formation of pits with a diameter of approximately 80 μm on the *Surveyor III*, positioned approximately 155 m from the landing site, as seen in Figure 5 below. J. Lane et al. (2008) found that around 2600 kg of lunar soil was ejected during this landing, illustrating the plume's extensive reach and its capacity for surface modification.

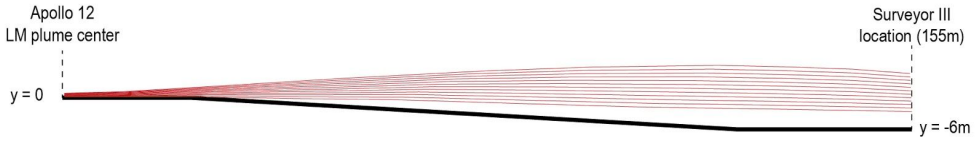


Figure 5: The main ejecta sheet (dust, sand, gravel, rocks) travels upward at a 1-3 degree slope away from the landing spacecraft. *Surveyor III*, located beneath this sheet, was partially shielded from impacts. Some ejecta scatter into higher and lower angles. Provided by Tosh et al. (2012).

Details concerning the erosion rate equation are provided in **Appendix** B for clarity.

2.1.2 Dissimilarities of Plume-Surface Effects on Mars

The attenuated atmosphere of Mars exerts a pivotal influence on the behaviour of exhaust plumes. Contrasting with the lunar environment, where the vacuum permits unrestrained plume expansion, the Martian atmosphere facilitates the collimation of exhaust, thereby concentrating the plume more acutely as it impacts the surface. Philip Metzger, J. Lane, et al. (2021) and Allen et al. (2012) have underscored that this *collimation* engenders a denser plume with augmented total pressure on the Martian surface, consequently intensifying erosion rates and inducing more pronounced topographic modifications compared to lunar conditions. The atmospheric confinement on Mars results in elevated localised impingement pressures, which lead to substantial erosion and the formation of deeper craters, a phenomenon extensively investigated by M. Mehta, Sengupta, et al. (2013). Empirical analyses of human-sized landers on Mars indicate that significant cratering will ensue beneath the exhaust plume, with soil and gravel being ejected into high trajectories, thus creating a blast zone approximately 1 kilometer in radius, as noted by P. T. Metzger, Li, et al. (2009). This scenario starkly contrasts with lunar conditions, wherein the vacuum environment permits exhaust gases to disperse more freely, thereby mitigating the intensity of surface interactions.

Intriguingly, studies by M. Mehta, Renno, et al. (2011) corroborate these observations, elucidating that the less cohesive nature of Martian soil results in elevated erosion rates, a phenomenon Greeley et al. (2010) observed at altitudes as high as 63 meters, an occurrence not noted on the Moon. The erosive processes on Mars not only amplify the impact but also significantly extend the affected area, as M. Mehta, Renno, et al. (2011) demonstrated through experiments at the (Ames Research Center) Planetary Aeolian Laboratory (PAL), where transient craters up to 1 meter in diameter formed in under a second. During the Phoenix mission, for instance, Rennó et al. (2009) detailed that surface patches of soil were cleared by the lander's 12 engines, which pulsated at 10 Hz with an average impingement pressure only one-tenth of Earth's sea-level pressure. This pulsation revealed subsurface ice several centimeters below the surface, over a radius of 75 to 85 cm, suggesting that the engine plume may have induced deep cratering in the Martian regolith, a phenomenon not typically observed during lunar landings, as noted by M. Mehta, Renno, et al. (2011). Additionally, unlike the Moon's extremely compacted sub-surface, which offers a stable landing surface, the Martian soil is continuously disturbed by wind and volatile cycles, rendering it less compacted and more susceptible to erosion. These wind and volatile cycles, illustrated in Figure 6 below, are commonly known as *"Dust Devils"*, atmospheric phenomena formed by the interaction of rising warm air and descending cooler air. These *Dust Devils* can become significantly large and are most active during the spring and summer months in Mars' northern hemisphere. The challenge in the context of PSI lies in predicting their occurrence at specific locations, a task that remains elusive for scientists.

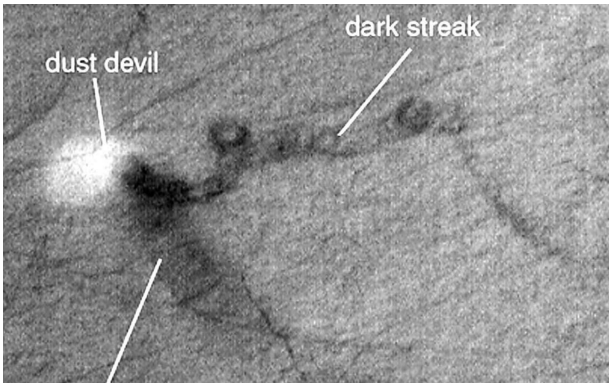
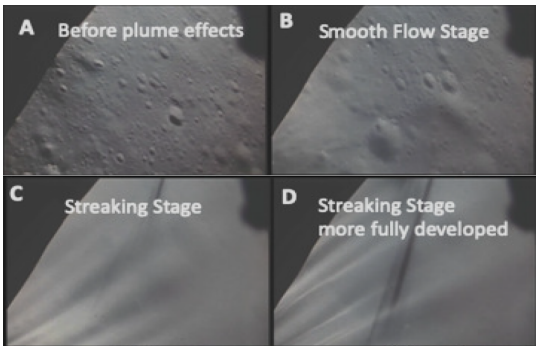


Figure 6: Dust Devil Image Captured by Mars Global Surveyor. Under 100 m Wide, Within a 1.5 by 1.7 km Area. Provided by Planetary Data System (2022).

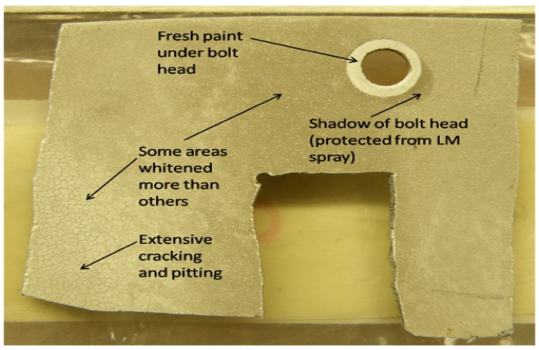
An aspect that, surprisingly, often eludes scholarly attention yet retains substantial importance is the influence on the thermal environment. Leovy (2001) observed temperature values on Mars ranging from −143 °C to 35 °C, a range less extreme compared to the Moon's temperature fluctuations, which span from −183 °C to 127 °C, as documented by Williams et al. (2017). Farries et al. (2021) argues that a comprehensive understanding of the regolith's properties is paramount for ensuring the safe traversal to and from the Moon and Mars. The continuous alteration or degradation of the regolith by extreme temperature fluctuations, for instance, poses significant challenges to the reliability of these understandings. Martian regolith are generally smaller, where around 2 μm in radius has been observed, affecting visibility and the potential for hardware abrasion differently due to the planet's atmospheric interaction. This difference in particle size and behaviour leads to distinct deposition patterns, where ejected particles on Mars, as noted in *Section 1.1*, tend to form larger and more dispersed clouds, differing significantly from lunar conditions.

2.1.3 Operational Consequences of PSI

The operational ramifications of PSI present formidable and analogous challenges for spacecraft during landings on both *lunar* and *Martian* terrains, despite the distinct environmental conditions characteristic of each celestial body. To minimise redundancy, their impacts on spacecraft operations will be examined concomitantly. Notably, NASA (2020) and International Agency Working Group (2016), among others, have extensively documented issues such as the destabilisation of Lander Module's, damage to exposed hardware, and severe visibility reductions. Figure 7 (a) below illustrates the latter issue, depicting the progressive decrease in visibility of the *Apollo 15* Lunar Excursion Module (LEM) camera from positions A to D as the spacecraft approached the lunar surface. This camera, essential for subsequent analytical assessments on Earth, became increasingly ineffective, a concern previously underscored by Mitchell et al. (1971).



(a) Apollo 15 LEM camera captures show the development of exhaust plume and regolith interactions, leading to rapid particle sheet formation that obstructs the view. Adapted from P. T. Metzger (2020).



(b) Surveyor III Impact. Provided by Watkins et al. (2021).

Figure 7: Manifestations of Regolith Interactions on Lunar Landing and Proximity Equipment.

Expulsion of regolith dust by landing modules, as documented by Colwell et al. (2007), introduces *ultra-fine* particles, typically tens of microns in diameter into the environment. These particles are capable of adhering to and penetrating sensitive equipment, such as the joints of Extravehicular Activity (EVA) suits—-a problem, predominately, noted during the *Apollo* missions. Immer et al. (2011) detailed the return of *Surveyor III*, as previously referred to in *Section 2.1.1*. Components provided concrete evidence of significant dust accumulation and physical abrasion at a distance of 155 m from the *Apollo 12* landing site, with notable impacts such as cratering and pitting on its surface. Figure 7 (b) reveals fresh paint beneath the bolt head, which was shielded from LM spray, maintaining its original appearance. In contrast, the surrounding exposed areas display extensive cracking, pitting, and a notable whitening effect that varies in

intensity, a stark reminder of the duality of Plume Surface Interactions. Moreover, the excavation and displacement of surface material directly beneath the landing spacecraft can create an unstable and uneven surface, potentially compromising the mission's success. Such irregular terrain, as Rahimi et al. (2020) explored, may cause the spacecraft to rest at an angle, placing sensitive propulsive or scientific equipment dangerously close to the surface. A notable instance of this hazard occurred during the NASA *Apollo 11* Mission, where the proximity of the Lunar Excursion Module (LEM) to the crater rim resulted in the structural failure of its descent engine nozzle skirt. The plume flow from the descent engine back-filled and over-pressurised the nozzle due to its position near the crater edge.

The interaction between exhaust plumes and lunar regolith engenders high-velocity dust ejection, presenting a substantial hazard to optical instrumentation. Christoffersen et al. (2008) noted that this dust, characterised by sharp-edged, irregularly-shaped particles, can inflict considerable abrasive damage upon impact with surfaces. Specifically, during the *Apollo* missions, efforts to remove this pervasive dust led to severe abrasion of displays and visors, significantly impairing their functionality. The deleterious effects of regolith extend beyond stationary equipment to mobile units, such as rovers. Both lunar and Martian rovers have encountered operational impediments due to dust accumulation. As delineated in Table 14 in **Appendix** E, dust accumulation and the resultant clogging of equipment presented significant challenges during the *Apollo 17* mission, making up over 50% of all hazards recorded. Furthermore, Y. Wang (2014) posited that likely failure of the Yutu lunar rover's thermal control systems was precipitated by dust interference. Additionally, Leger et al. (2005) determined that the Spirit Mars rover experienced wheel failures attributable to dust infiltration in the bearings. The Mars Science Laboratory's employment of the sky crane landing technique is an innovative approach to mitigating PSI effects. By maintaining the thrusters elevated above the ground, the landing platform minimised Martian disturbance. Nonetheless, P. T. Metzger, Li, et al. (2009) illustrated that even this technique could not entirely obviate the issue, evidenced by the damage to an anemometer on the Curiosity rover caused by ejecta during its landing.

2.1.4 Physics and Flow Structure of Supersonic Plumes

When a supersonic flow is discharged into an environment where the ambient pressure is lower than that within the flow, an *underexpanded jet* is produced (Franquet et al. 2015). The resultant pressure disparity necessitates the flow to expand upon exiting the nozzle to achieve equilibrium with the surrounding pressure. This pressure equalisation is prominently observed at the jet boundary, which functions as an interface where the pressures on either side must converge (Cai et al. 2016). The ensuing expansion, characterised by the intricate interplay of compressible and viscous forces, delineates the flow's architecture. Proximal to the nozzle, a stable potential core emerges, governed primarily by compressibility, with viscous effects being negligible (Radtke et al. 2019). This core undergoes isentropic expansion followed by *recompression* through a series of internal compression waves. As illustrated in Figure 8 below, the compression and *expansion* waves near the nozzle are distinctly visible, demarcating the regions where the flow adjusts to the ambient pressure. Adjacent to the core is the mixing layer, where viscosity mitigates the velocity discontinuity at the jet boundary (Panda and Seasholtz 1999). This region is pivotal in configuring the jet profile, as viscous forces instigate instabilities that augment entrainment, thereby broadening the mixing layer and the jet's diameter. The mixing layer itself encompasses a faster-moving supersonic inner region and a slower-moving subsonic outer region. This is depicted by the shear layer/mixing layer, demonstrating the initiation of the mixing process near the nozzle and its extension downstream (Wu et al. 2019).

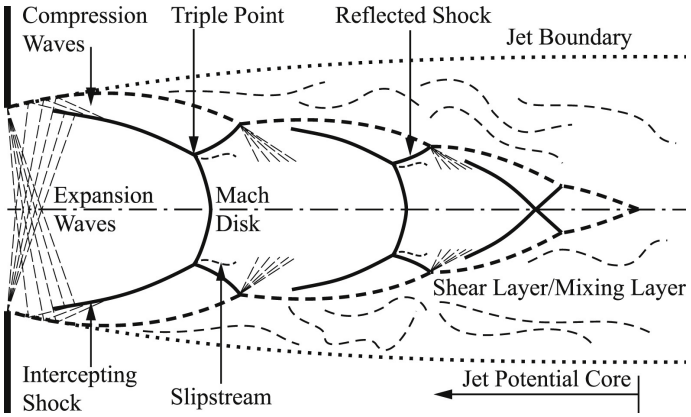


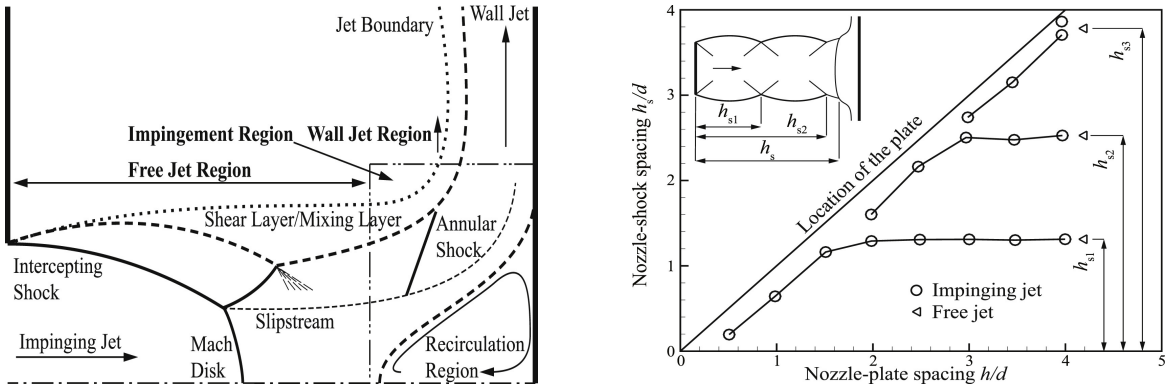
Figure 8: Scheme of a Moderately Under-Expanded Free Jet. Provided by Jiang et al. (2019).

The dynamics of the mixing layer are fundamentally governed by the Reynolds number, with turbulent conditions prevailing when the Reynolds number exceeds 10^4 (Franquet et al. 2015). This turbulence propagates linearly from the nozzle, significantly contributing to the jet's spreading. As depicted in Figure 8, the turbulent mixing layer broadens as it progresses downstream, illustrating the intensified mixing and entrainment processes resultant from the high Reynolds number. At elevated pressure ratios (specifically for air, when the ratio surpasses approximately 2.1), the jet undergoes such extensive expansion that the pressure along the central axis becomes insufficient for a simple *oblique shock* to recompress the flow (Henderson 2002). Consequently, a normal shock, known as a Mach disk, forms. This *Mach disk*, as depicted in Figure 8, serves as a pivotal juncture where the intersecting shock is reflected, converging with all three shock types at a location termed the *triple point* (Panda and Seasholtz 1999). Downstream of the Mach disk, the flow transitions to subsonic speeds, while remaining supersonic outside this boundary adjacent to the reflected shocks. This dichotomy generates a slipstream, delineating two concentric flow regimes with equivalent pressure and direction but disparate velocities. Figure 8 clearly illustrates this transition, showing the Mach disk, reflected shocks, and the slipstream that separates the supersonic and subsonic regions. As the pressure ratio increases, the subsonic core is rapidly reaccelerated to supersonic speeds, perpetuating the shock structure pattern. However, at even higher pressure ratios, a singular, dominant shock cell emerges, precluding the formation of subsequent normal shocks. Following these intense normal shocks, the jet decelerates through a succession of weaker *oblique shocks*, though these are often distorted by flow instabilities, ultimately leading to a conventional subsonic deceleration in the far field (Franquet et al. 2015). This behaviour is depicted in Figure 8, showcasing the successive shock structures and their downstream evolution, highlighting the complex interplay of shocks within the flow. Figure 8 also elucidates how the intersecting shock and slipstream contribute to the overall flow stability and mixing efficiency. The expansion waves and their interaction with the jet boundary underscore the challenges inherent in managing such flows, particularly in varied planetary environments. This comprehensive understanding is imperative for developing effective mitigation strategies for PSI, as it elucidates the fundamental physical processes governing plume behaviour.

2.1.5 Physics of Jet Impingement

The formation of shock patterns in an impinging jet is governed by a multifaceted interplay of various parameters, extending beyond the outlet-to-ambient pressure ratio that predominantly characterises free jets (Kandakure et al. 2008). To comprehensively delineate the flow field structure, it is imperative to consider additional factors such as the Mach number proximate to the impingement zone, the impingement angle, and the normalised nozzle-to-wall distance relative to the nozzle exit diameter (Franquet et al. 2015). The intrinsic complexity of the flow precludes its classification based on a solitary parameter, necessitating a more nuanced and multifactorial approach. Conventionally, the *underexpanded* impinging jet is segmented

into three distinct regions: the free jet region, the impingement region, and the radial wall-jet region (Nakai et al. 2006). As illustrated in Figure 9 (a), the flow field pattern of a supersonic jet emanating from an axisymmetric round nozzle exhibits similarities to free jets in the near-field, with structures in this region being predominantly influenced by the nozzle's operational conditions and pressure ratio (Lamont and Hunt 1980). Furthermore, under fixed operational conditions, the length of the fully developed shock cells and the stand-off distance remain largely insensitive to variations in the impingement distance, as depicted in Figure 9 (b) (Risborg and Soria 2009).



(a) Impinging Supersonic Jet Scheme.

(b) Positions of Shock Structures at Varying Impingement Distances.

Figure 9: Impinging Flow Structure. Both provided by Jiang et al. (2019).

Upon impinging upon the wall, the flow transitions into the impingement region, characterised by substantial pressure gradients induced by the presence of the wall (Kalghatgi and Hunt 1976). These gradients necessitate a dramatic alteration in flow direction, with the extent of the interaction in this region contingent upon the nozzle-plate distance (Nakai et al. 2006). A closer proximity of the potential core to the impinged surface results in more pronounced interactions. Depending on the Mach number, impingement distance, and the degree of flow underexpansion, a detached shock or *recirculation region* may develop within the impingement region. In moderately underexpanded flows, the flow downstream of the oblique shock reflection on the axis remains supersonic, often culminating in the formation of a detached shock orthogonal to the jet flow, referred to as the *stand-off* or plate shock. This shock serves to equilibrate the pressure near the surface (Kandakure et al. 2008). Conversely, when the shock cell proximal to the plate contains a Mach disk, it induces a higher total pressure loss compared to the reflected shock, potentially resulting in an off-axis pressure peak along the plate. This pressure peak, posited to be generated by the impingement of the slipstream (Figure 9 (a)), may lead to the formation of a conical-shaped stagnation bubble in front of the plate (Lamont and Hunt 1980). However, these bubbles have been shown to exert minimal influence on the subsequent development of the wall jet region. Despite numerous studies detecting such recirculation regions, the precise conditions under which they form remain elusive. In the presence of a Mach disk, the reflection of the expansion fan as compression waves on the slipstream between the supersonic and subsonic flow ultimately coalesce, forming an annular shock (Risborg and Soria 2009). The final region, known as the wall jet region, is characterised by the emergence of the wall jet. In this region, the jet moves radially outward, undergoing an expansion that can re-accelerate the flow to supersonic speeds. Under these conditions, shock cell trains resembling the quasi-periodic structures in a free jet may develop along the wall jet (Henderson 2002). The shear layer on the upper surface of the wall jet and the boundary layer that forms along the wall eventually merge, reducing the flow to subsonic, akin to the behaviour observed in free jets.

2.2 Mitigating and Controlling PSI

2.2.1 Existing Proposals

The evolving landscape of *lunar* and *Martian* missions underscores the imperative for efficacious mitigation strategies to manage PSI during spacecraft landings and launches. Vital to safeguarding both the spacecraft and the surrounding infrastructure from the erosive effects of exhaust plumes, which can dislodge and propel regolith at elevated velocities, thereby posing significant risks to mission success and safety. Among the proposed methodologies for managing plume-surface interactions during landing, the deployment of erosion-resistant landing pads is frequently regarded as the most optimal approach. This consensus arises from the high TRL associated with landing surface methods compared to other mitigation techniques (Watkins et al. 2021). These methods typically involve the construction of landing pads utilising locally available or readily manufacturable materials, such as regolith bricks or Martian concrete. A notable advanced landing surface construction technique is regolith sintering, which employs solar energy to thermally bond surface regolith material. This approach demands minimal civil engineering infrastructure among the proposed landing pad construction techniques. As discussed by Lin et al. (1997) and Meurisse et al. (2017), regolith sintering leverages the abundant solar energy on these celestial bodies to fuse lunar or Martian soil into a solid structure capable of withstanding the forces exerted during landing and takeoff. This method is particularly advantageous as it minimises the necessity for transported materials, thereby enhancing its practicality and feasibility for extraterrestrial applications.

However, the construction of landing pads on extraterrestrial surfaces presents significant environmental challenges. T. L. Wilson and K. B. Wilson (2005) observed that the regolith sintering process irrevocably alters the landscape, thereby compromising the scientific integrity of geological surveys and the conservation of pristine extraterrestrial terrains. Duri et al. (2021) underscored these concerns, emphasising their critical importance for the continued advancement of space exploration, as such alterations could obscure planetary history and affect undetected ecosystems. Consequently, a cautious approach to regolith sintering is imperative to preserve the natural state of celestial bodies for future scientific research and exploration. Moreover, Farries et al. (2021) elaborated that these technologies remain in the developmental stages, necessitating further research to validate their viability, reflected by a low TRL score of 3. Implementation of surface-based techniques also involves substantial civil engineering challenges. For instance, the construction of regolith bricks or the laying of lunar concrete, as examined by R. P. Mueller et al. (2016) and G. Davis et al. (2017), demands significant preparatory work and machinery, which must be transported and operated under harsh planetary conditions. While the creation of landing pads mitigates PSI, it does not entirely protect nearby scientific instruments from the scouring effects of redirected plume flow.

To enhance these protective measures, newer proposals, such as the formation of barrier walls around landing sites, have been suggested by Hintze and Quintana (2013) to shield critical structures from the abrasive forces of landing plumes. Nonetheless, Kelso et al. (2016) and others have pointed out that adapting these strategies to different planetary environments remains a complex endeavor. The International Agency Working Group (2016) report echoes this sentiment, stressing that the efficacy of a given technology in one context does not guarantee its success in another, thereby highlighting the necessity for a suite of adaptable and robust technologies. The implementation of spacecraft-based mitigation techniques could obviate the infrastructure-centric operational challenges intrinsic to surface-based mitigation strategies, although few such methods have been conceptualised and developed to date. The operational complexities associated with both surface-based and spacecraft-based techniques underscore the persistent challenges in PSI mitigation. The necessity for preliminary missions to prepare landing sites or the deployment of autonomous machinery to construct requisite infrastructure exemplifies the substantial logistical obstacles that remain. P. T. Metzger (2020) corroborates these findings, asserting that no single technology can comprehensively address the challenges posed by lunar and Martian dust. Rather, an amalgamation of strategies, tailored to mission requirements and environmental conditions, will likely be requisite to effectively mitigate these pervasive issues.

2.2.2 Synthetic Jet Actuators: Microjets

Building upon the comprehensive analysis of existing research, the use of microjets stands as a promising solution for mitigating PSI during spacecraft landing operations on extraterrestrial bodies. Although this strategy remains relatively *niche* compared to other forms of flow control methods, it offers unique advantages by leveraging the fundamental physics of boundary layer and shear layer interactions, as well as vorticity generation and pressure redistribution. Microjets introduce localised high-velocity streams into the boundary layer of the primary jet, destabilising it and promoting an early transition to turbulence. This transition enhances mixing and increases the energy dissipation rate, thereby reducing peak thermal and mechanical loads on the impingement surface. The turbulent boundary layers fostered by microjets are more effective at mixing, leading to a more uniform energy distribution that mitigates localised thermal and mechanical stresses. This capability is crucial for preserving the integrity of both the landing surface and the spacecraft. Additionally, the primary jet emanating from a rocket nozzle generates strong shear layers that define the core of the plume. Strategic introduction of microjets disrupts these shear layers, causing a more uniform spreading of the jet. This modification helps diffuse the kinetic energy of the primary jet over a larger area, thus reducing localized impact forces and heating on the landing surface. Such an approach is vital for minimizing surface erosion and structural damage during landing.

Moreover, microjets generate additional vortical structures within the primary jet flow, which enhance the entrainment of ambient gases. This increased entrainment promotes effective mixing and diffusion rates, leading to a more distributed pressure profile. As a result, the concentrated impact that typically results in high-pressure and thermal loads is mitigated, preserving the surface and reducing the risk of material ejection. Furthermore, by injecting microjets at angles to the main flow, the resultant deflection can steer the primary jet flow away from critical surfaces. This steering effect creates a more favorable pressure distribution on the landing pad or surface, reducing the intensity of plume impingement. The strategic placement of microjets ensures that the primary jet's impact is distributed more evenly, decreasing the likelihood of localized surface damage and improving overall landing stability.

The application of microjets for PSI mitigation provides a spacecraft-based solution that can be integrated with existing propulsion systems, offering a flexible and effective means of reducing the destructive effects of plume impingement. The benefits of this approach include enhanced mixing and turbulence, shear layer disruption, vortical structure formation, and flow deflection. These advantages collectively contribute to a more distributed pressure profile, reduced peak thermal and mechanical loads, diminished localised impact forces, and improved surface integrity. By introducing controlled disturbances into the boundary and shear layers of the primary exhaust plume, microjets enhance mixing, disrupt shear layers, and redistribute pressure and velocity fields. This innovative approach promises significant improvements in landing safety and surface preservation, which will be further tested in the subsequent analysis.

2.3 Literature for the Development of Numerical Methodologies

In the process of devising a method for simulating nozzle flow, we conducted a thorough analysis of the seminal work "Design of a Supersonic Nozzle" by McCabe (1967). Although McCabe's research primarily emphasises wind tunnel experiments, subsequent investigations by Davies (2023) into Martian atmospheric conditions have revealed analogous flow structures. These include the presence of a pronounced bow shock, a distinct shear layer, and a triple point that gives rise to tail shocks and wall jets. Despite certain variations attributable to the unique Martian environment, the fundamental similarities in flow dynamics between terrestrial and Martian conditions substantiate the application of McCabe's design principles in our validation process.

2.3.1 "Design of a Supersonic Nozzle" by McCabe (1967).

The supersonic nozzle was originally conceptualised by McCabe (1967) through the application of then-state-of-the-art rigorous numerical methods. His extensive work employed the complete Method of Characteristics (MOC) approach, ensuring continuous nozzle curvature by specifying a continuous Prandtl-Meyer (PM) angle gradient along the nozzle axis. This design process resulted in a boundary-layer-corrected supersonic nozzle geometry for Mach 5 exit flow, characterised by its minimal length and optimal operational parameters throughout its length. A distinctive feature of McCabe's design was the provision of precise x-y coordinate data corresponding to the nozzle profile, a feature that is seldom available in nozzle design literature. Originally intended for supersonic wind tunnel testing, this nozzle design facilitated an exit velocity conducive to aerospace applications. The schematic, in Figure 10, illustrates McCabe's final nozzle geometry, emphasising its key characteristics and nozzle streamline, although the depicted curvature primarily serves to illustrate the design methodology rather than the exact published geometry. For further reference, the two-dimensional coordinate data from McCabe's work is reproduced in **Appendix** C.

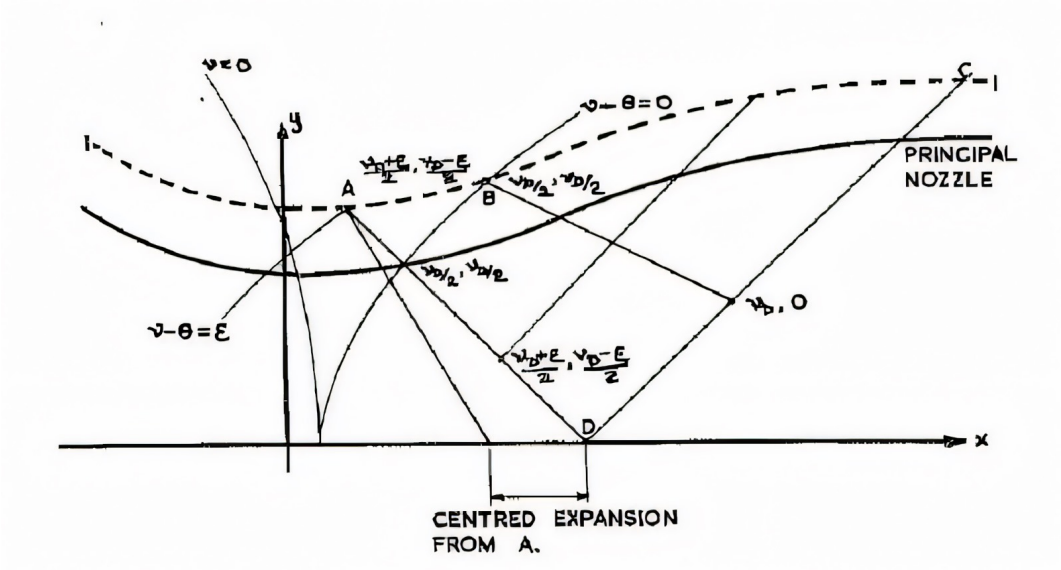
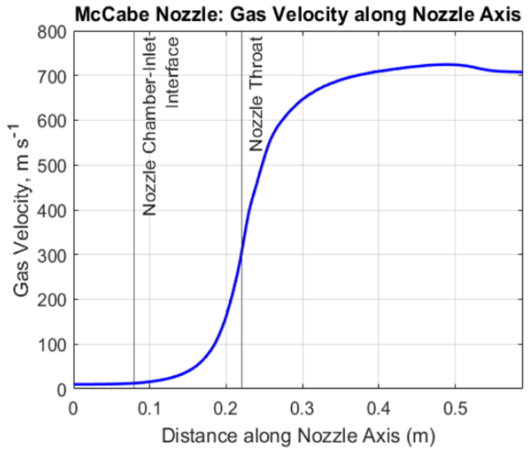
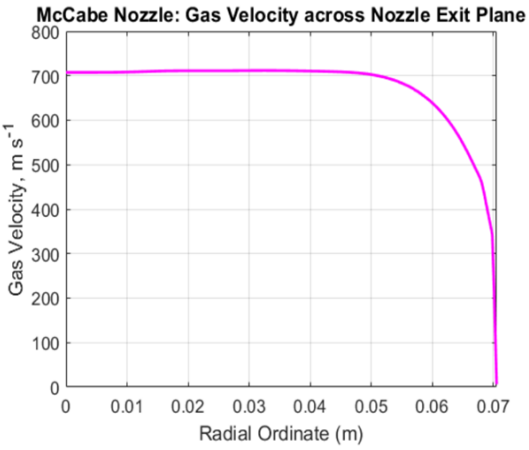


Figure 10: Supersonic C-D Nozzle Geometry. Provided by McCabe (1967).

Validating a numerical simulation method developed from McCabe's design necessitates comparing the simulated nozzle flow against the published design flow properties, such as the exit Mach number, and ensuring uniform acceleration and exit flow uniformity across the nozzle diameter, as corroborated by existing literature on supersonic nozzles. Building upon the technical groundwork laid by McCabe (1967), our research has adeptly integrated these methodologies into our experimental designs, specifically utilising the detailed x-y coordinate data and computational strategies. The adaptation of McCabe's methodologies, as highlighted by Davies (2023), was pivotal for our subsequent work. By employing these methods, we configured our nozzles for hypersonic tests, ensuring a robust experimental setup. This integration not only facilitated direct comparisons between our experimental results and those derived from Davies' methodologies but also significantly enhanced our validation process, thereby underpinning the development of our numerical methodology. Davies (2023) demonstrated the characteristics of a supersonic nozzle designed using McCabe's methodologies. The gas velocity along the nozzle axis showed a uniform acceleration from the nozzle throat to the exit plane, as seen in Figure 11 (a). The acceleration rate was predominantly linear, particularly near the nozzle throat, with a plateau towards the nozzle exit likely due to operating under off-design conditions using a Martian atmosphere model. Figure 11 (b) shows the gas velocity across the nozzle exit plane was also uniform, with a mean gas velocity corresponding to a Mach number of approximately 5.1, close to the expected value of 5.0 indicating effective nozzle performance under the specified conditions.



(a) 2D McCabe Nozzle Simulation: Gas Velocity along Nozzle Axis.



(b) 2D McCabe Nozzle Simulation: Gas Velocity across Nozzle Exit Plane.

Figure 11: 2D McCabe Nozzle Simulation Results. Both provided by Davies (2023).

Additionally, Figure 12 illustrates the flow structures at the point of impingement included a strong bow shock, a clear shear layer, and a triple point leading to well-defined tail shocks and radially propagating wall jets. These features are characteristic of an underexpanded jet and were captured accurately in the simulation, validating the methodologies used. Finally, the Mach contours obtained, depicted in Figure 13 below, shows the Mach number distribution along the flow field. The image highlights significant flow features, such as the high Mach number core exiting the nozzle, the mixing layer, and the formation of shock structures. The smooth gradient of Mach numbers and the clear delineation of flow regions indicate that the flow behaviour matches theoretical expectations for an *underexpanded jet*. The Mach contours visually confirm the presence of critical flow phenomena such as shock diamonds and expansion fans, providing further validation of the simulation accuracy. These results serve as a crucial reference for comparing our experimental outcomes, ensuring that our methodologies align with the established benchmarks set by Davies work.

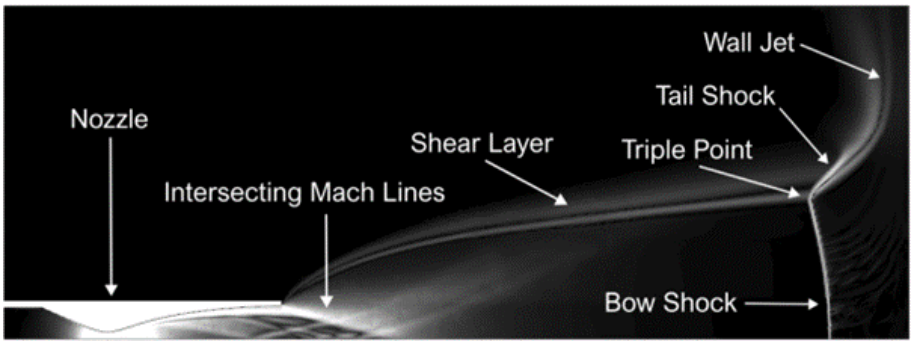


Figure 12: 2-D Simulation of McCabe Nozzle Flow, Pseudo-Schlieren Image. Provided by Davies (2023).

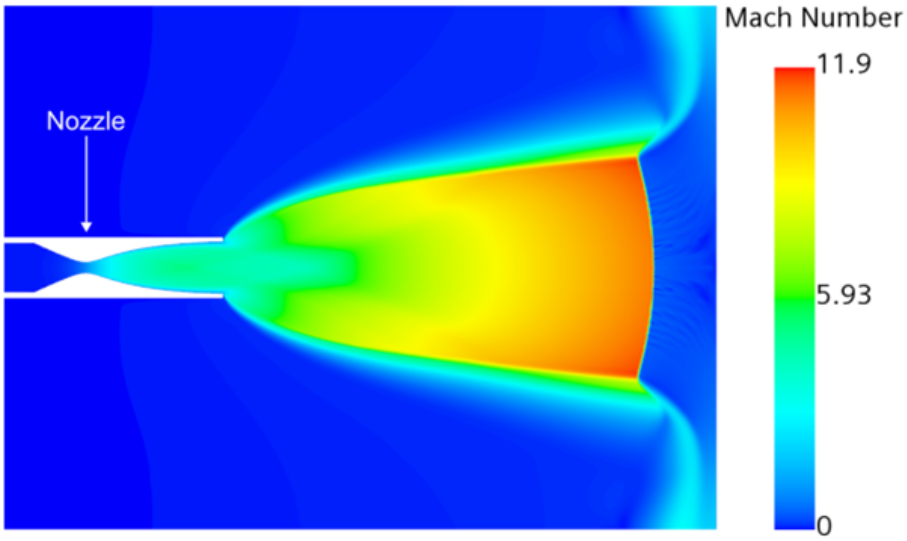
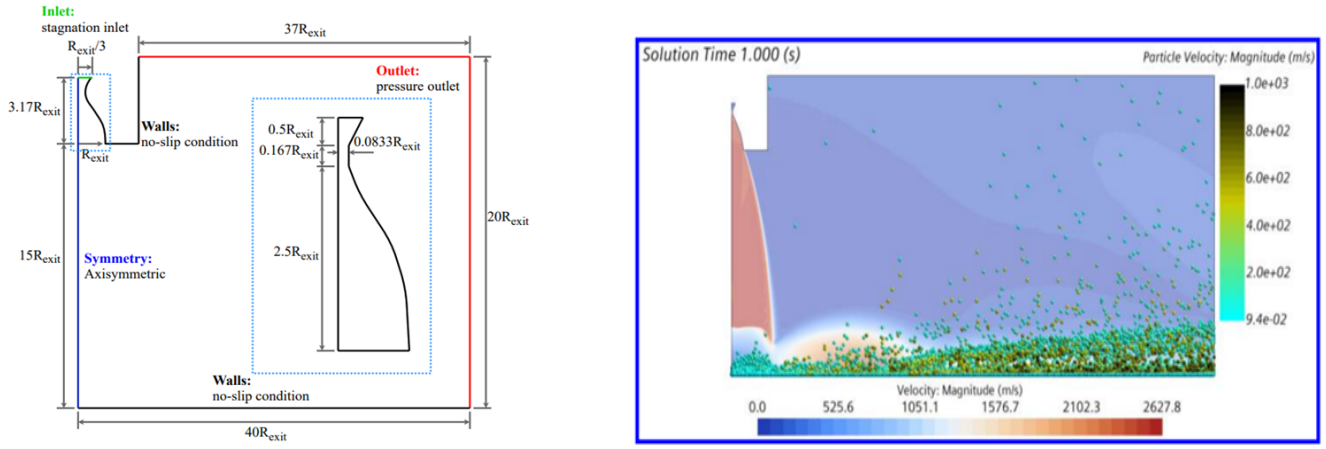


Figure 13: 2-D Simulation of McCabe Nozzle Flow, Mach Contours. Provided by Davies (2023).

2.3.2 "Rocket Plume Interacting with Mars Soil Particulates" by Fontes and Philip Metzger (2022): Integration of Nozzle Design

Fontes and Philip Metzger (2022) conducted an exhaustive numerical investigation to elucidate the interactions between a rocket plume and the Martian surface. Although their primary focus was the behaviour of soil particulates entrained by the exhaust plume, the study also furnished valuable insights into the rocket nozzle design and the computational domain utilised for the simulations. The physical domain, as delineated in Figure 23 (a) overleaf, was constructed to represent a rocket nozzle with an exit radius of 0.6 m, situated at a height of 15 nozzle exit radii (9 m) above the ground surface. The computational domain extended radially 24 m from the axi-symmetric centre and included a total height of 40 nozzle exit radii (24 m). This domain was designed to capture the complete extent of the plume's interaction with the Martian surface, including the radial expansion of the flow post-impingement. Furthermore, the nozzle geometry was engineered to generate sufficient thrust to sustain a 32.5-ton lander in a hovering state within the Martian environment. The converging section of the nozzle compressed the hot gas at 3300 K as it approached the throat, where it attained sonic velocity. Subsequently, the flow expanded through the divergent section of the nozzle into the Martian atmosphere, forming an *underexpanded* gas jet. The total stagnation pressure at the inlet was set to 200 times atmospheric pressure (101,325 kPa) to ensure optimal nozzle performance.

The computational domain was discretised utilising a mesh composed of trimmed quadrilateral elements. The researchers implemented second-order advection schemes for the momentum and turbulence transport equations, and the balance equations were discretised employing the finite volume method. The high-velocity flow was managed using the advection upstream splitting method plus (AUSM+), which is particularly adept at capturing contact discontinuities, normal shock discontinuities, and ensuring smooth transitions through the sonic point. Although the study principally concentrated on the behaviour of soil particulates entrained by the exhaust plume, the numerical simulations furnished valuable insights into the flow dynamics within the nozzle and the near-field plume region. As depicted in Figure 23 (b) below, the velocity magnitude contours of the gas jet flow illustrate the high-speed flow exiting the nozzle and impinging upon the ground surface. The plume velocity reached values exceeding 1000 m/s near the surface, providing critical data for subsequent nozzle flow simulations.



(a) Nozzle 2 Domain and Nozzle Dimensions

(b) Nozzle 2 Velocity Contours

Figure 14: 2D McCabe Nozzle Simulation Results. Both provided by Fontes and Philip Metzger (2022).

The specified domain dimensions and resolution, alongside the nozzle design parameters, serve as foundational benchmarks for subsequent numerical studies investigating the impacts of various nozzle configurations, ambient conditions, and surface interactions. Therefore, while Fontes and Philip Metzger (2022) predominantly examined the behaviour of soil particulates entrained by the rocket plume, their research also furnished crucial insights into the design of the rocket nozzle and the computational domain employed for simulations. The exhaustive, and detailed, description of the nozzle geometry and the flow conditions within the domain proved vital for the subsequent analysis.

3 Numerical Methodology

3.1 Computational Fluid Dynamics

Numerical simulations within this study were conducted utilizing *Simcenter* STAR-CCM+, a comprehensive Computational Fluid Dynamics suite. The CFD tools resolved the governing compressible Navier-Stokes (NS) equations throughout the simulation domain. These equations, presented below in condensed symbolic notation and adapted from Schlichting and Gersten (2017), encapsulate the fundamental principles governing the flow dynamics within the context of the study:

Continuity Equation

$$\frac{D\rho}{Dt} = -\rho\nabla \cdot \vec{v} \tag{1}$$

Momentum Equation

$$\rho\frac{D\vec{v}}{Dt} = \vec{f} - \nabla p + \nabla \cdot \tau \tag{2}$$

The viscous stress tensor, τ, is defined as:

$$\tau = \mu\left(2\dot{\epsilon} - \frac{2}{3}\delta\nabla \cdot \vec{v}\right) \tag{3}$$

where δ is the Kronecker delta tensor, with $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$.

Energy Equation

The energy equation, derived from the system's temperature balance, is given by:

$$\rho C_p\frac{DT}{Dt} = \nabla \cdot (\lambda\nabla T) + \beta T\frac{Dp}{Dt} + \phi \tag{4}$$

Here, the coefficient of thermal expansion, β, is defined as:

$$\beta = -\frac{1}{\rho}\left(\frac{\partial\rho}{\partial T}\right) \tag{5}$$

The dissipation function, ϕ, is:

$$\phi = \nabla \cdot (\tau\vec{v}) - \vec{v} \cdot \nabla \cdot \tau \tag{6}$$

In these equations, ρ represents the plume density, t is the physical time, $\vec{v}$ denotes the plume velocity, $\vec{f}$ is the body force, p is the plume pressure, τ is the viscous stress tensor, μ is the dynamic viscosity, $\dot{\epsilon}$ is the rate of strain tensor, C_p is the specific heat capacity at constant pressure, T represents the plume temperature, λ is the thermal conductivity, β is the coefficient of thermal expansion, and ϕ denotes the dissipation function. Vector quantities are indicated with an arrow, fluxes are denoted with a dot, and total derivatives are indicated by the D/Dt notation.

Simcenter STAR-CCM+'s robust meshing tools and solver capabilities allow for detailed resolution of the flow field, particularly in capturing the critical phenomena such as shock waves and turbulence interaction typical in hypersonic flows. The software's capacity to efficiently handle complex boundary conditions and provide real-time simulation feedback makes it an excellent tool for the intricate analyses required in this study.

3.2 2D Boundary Layer-Corrected Mach 5 Nozzle

3.2.1 Nozzle Geometry

The two-dimensional x-y coordinate data delineating the nozzle profile, as originally documented by McCabe (1967), were gathered and plotted to identify any discrepancies that may have arisen during the manual transfer and conversion from inches to millimeters. These data, presented in **Appendix** C, consisted of only 57 coordinate points. According to Davies (2023), this proved inadequate for generating a smooth Computer Aided Design (CAD) model appropriate for Computational Fluid Dynamics simulations. To address this limitation, a modified Akima cubic interpolation method was employed, leveraging *Mathworks* MATLAB—a software widely recognised for its matrix-based numerical analysis capabilities (Akima 1970). This approach increased the coordinate points from 57 to 500, thereby enhancing the precision of the nozzle profile. The refined nozzle geometry is illustrated in Figure 15.

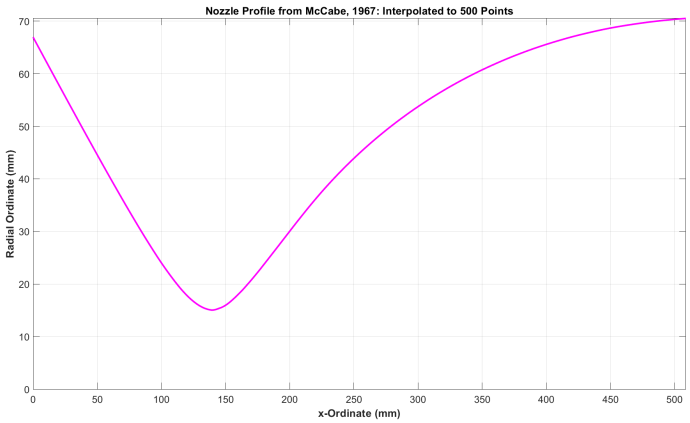


Figure 15: 2-D Boundary Layer-Corrected Mach 5 Nozzle: MATLAB Obtained Nozzle Contour.

Key nozzle characteristics, such as the radii and areas at the inlet, throat, and outlet, were calculated and are presented in Tables 1 and 2 below. The expansion ratio of the nozzle, defined as the ratio of the outlet area to the throat area, was determined to be approximately 22. Utilising *isentropic flow* equations, the exit Mach number was calculated to be approximately 4.84, providing critical data for the subsequent numerical flow analysis.

Table 1: McCabe Boundary-Layer-Corrected Mach 5 Nozzle: Nozzle Properties for Interpolated Geometry.

	Inlet	**Throat**	**Exit**
Area (mm^2)	14083.420	711.289	15617.983
Radius (mm)	66.954	15.047	70.508

Table 2: McCabe Boundary-Layer-Corrected Mach 5 Nozzle: Nozzle Properties for Interpolated Geometry.

Property	**Value**
Expansion Ratio	21.96
Nozzle Length (mm)	508.82
Exit Mach Number	4.84

The nozzle profile, derived using *Mathworks* MATLAB with 500 interpolated points, was systematically tabulated in a *Microsoft* Excel file. This data was subsequently imported into CAD software to generate the precise geometry of the nozzle, as depicted in Figure 16 overleaf. CAD tools were further employed to delineate the entire computational domain, strictly adhering to the prescribed dimensions, which will be elaborated upon in *Section 3.2.2*. The integration of this detailed profile into the CAD environment ensures an accurate representation of the nozzle geometry, crucial for high-fidelity CFD simulations (Wendt and Dick 2009).

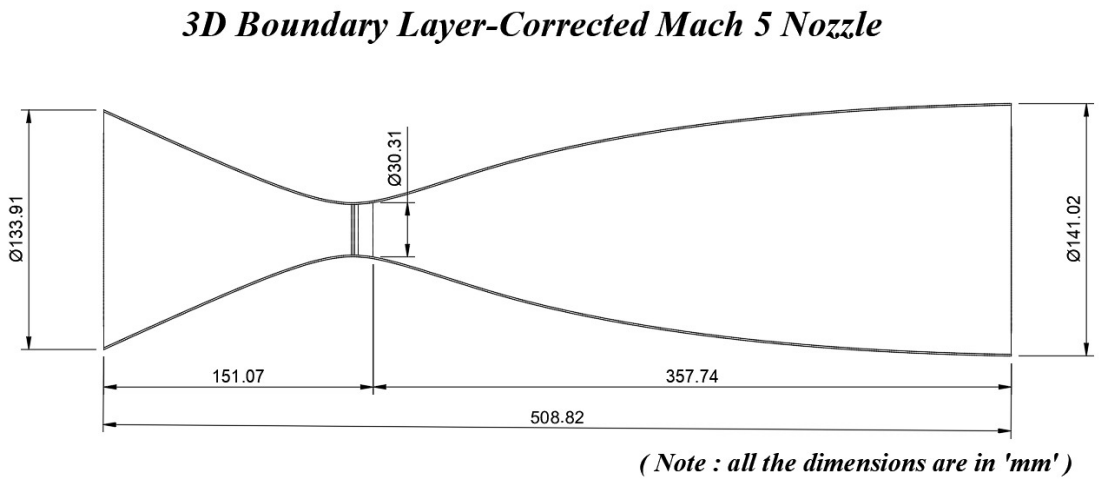


Figure 16: 2-D Boundary Layer-Corrected Mach 5 Nozzle: CAD Drawing.

3.2.2 Computational Mesh

The Nozzle CAD geometry was derived from the interpolated x-y coordinate data, as mentioned in *Section 3.2.1* , which was utilised to generate the solid body geometry, forming the basis of the synthesised computational mesh. Following the methodology of Davies (2023), an arbitrary dimension of $10d$ was prescribed for the distance between the nozzle exit plane and the impingement plate, as well as $5d$ for the height of the impingement plate, where 'd' represents the nozzle exit diameter used as a geometric normalising parameter. To maximise computational efficiency, planar symmetry was leveraged by aligning the nozzle axis with the domain x-axis, allowing for a half-height impingement plate. The overall domain dimensions, including the nozzle body, were set to 13.61 x $5.0d$ (*length* x *height*), as described by Davies. Additionally, a small chamber of arbitrary length $0.5d$ was included in the design.

The mesh development process was conducted iteratively, beginning with a coarse 2-D polygonal surface mesh and progressively incorporating regions of refined mesh to accurately capture anticipated flow effects. The initial base size for the mesh was set at 7.0 *mm*, with the cell surface growth rate reduced to 1.07 to ensure smooth transitions between varying cell sizes, enhancing the overall quality and fidelity of the mesh. While default mesh controls were retained for other parameters, specific adjustments were made to tailor the mesh to the study's requirements. Given the expectation of critical flow features along the plume length, at the impingement site, and at the interface between these regions, special attention was paid to these areas (Inman et al. 2008; Abbett 1970; D. M. Edgington-Mitchell et al. 2014). Rectangular refinement zones with reduced mesh base size were strategically placed to develop a finer mesh in these critical regions, denoted by 'A', 'B', 'C', and 'D' in Figure 17 overleaf. This targeted approach ensured that the mesh was sufficiently fine to capture the detailed dynamics of the flow in these areas. Although automated mesh refinement tools were available, manual refinement was preferred as it allowed for greater precision and control over the mesh quality in the regions of interest.

The general arrangement of the computational mesh is depicted in Figures 17 and 18 below, with geometry faces and refinement zones indicated.

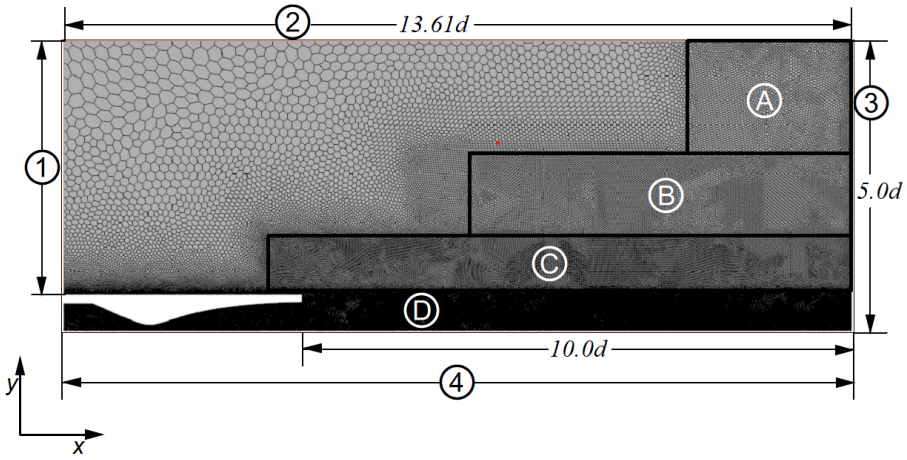


Figure 17: 2-D Boundary Layer-Corrected Mach 5 Nozzle Computational Mesh: Domain.

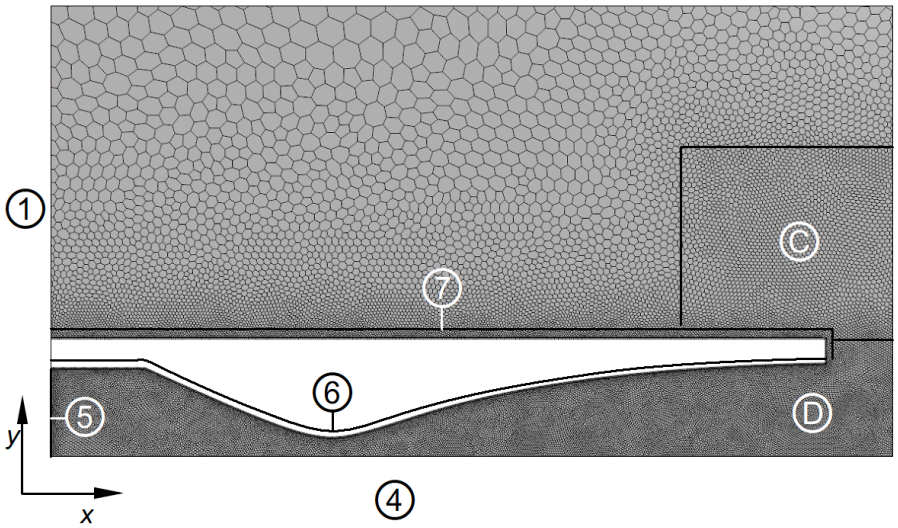


Figure 18: 2-D Boundary Layer-Corrected Mach 5 Nozzle Computational Mesh: Nozzle.

This approach not only optimised computational resources but also ensured that essential flow features were captured with high fidelity. The manual definition of refined mesh areas proved to be advantageous, providing a level of detail that automated tools could not consistently achieve. The resultant mesh was computationally efficient, striking a balance between fine cells in critical areas and coarser cells elsewhere, thereby maintaining computational efficiency without compromising on the accuracy of the simulation. The data in Table 3 overleaf correspond to the base mesh produced with the various components and mesh controls corresponding to Faces 1 - 7 seen in Figures 17 and 18 above. Additional mesh controls superimposed using regions of refinement A - D are detailed overleaf in Table 4.

Table 3: 2D McCabe Nozzle Flow Simulation: Base Mesh Controls for Faces 1-7.

Face(s)	Component	Target Surface Size (% of base size)	Minimum Surface Size (% of base size)	Prism Layer Controls
1	Free Stream	500	Default	Disabled
2	Free Stream	500	Default	Disabled
3	Impingement Plate	50	Default	Default
4	Symmetry Plane	n/a	n/a	n/a
5	Nozzle Inlet	15	7.5	2 Layers; Default Total Thickness
6	Nozzle Internal Surfaces	15	7.5	2 Layers; Default Total Thickness
7	Nozzle External Surfaces	20	10	Default

Refinement Zone	Target Surface Size (% of base size)
A	90
B	75
C	50
D	25

Table 4: 2D McCabe Nozzle Flow Simulation: Rectangular Refinement Zones.

3.2.3 Physics Models and Boundary Conditions

Symmetry was utilised to enhance computational efficiency, necessitating the adoption of a two-dimensional axisymmetric spatial model. To address the insufficient chamber length of only $0.5d$, numerical enforcement of uniform flow was implemented, ensuring adequate flow development. Optimal convergence in the simulation was achieved using an implicit unsteady time-stepping scheme, which employed coupled energy and flow solvers. To simulate Martian conditions for potential future applications, the simulation properties were adjusted to represent the Martian environment. This included a reduced-order model of the Martian atmosphere at low altitude, consisting of 2% argon (Ar), 2% nitrogen (N_2), and 96% carbon dioxide (CO_2) developed by K. Davis et al. (2015), with gas components assumed to be non-reactive and ideal. Initial domain physics conditions were derived from scaling parameters between Earth sea-level and Mars surface atmospheric conditions. The plume flow was modeled as pure nitrogen to ensure the correct ratio of specific heats, denoted as γ, between the gas plume and the ambient atmosphere. Given the typically turbulent nature of impinging flows, an appropriate turbulence model was applied. The standard Spalart-Allmaras turbulence model was chosen due to its widespread use in aerospace applications (Spalart and Allmaras 1992). Although specific inlet conditions for the McCabe nozzle were unavailable, Davies (2023) estimated these conditions by comparing them with a nozzle exhibiting a similar pressure ratio. Inlet supersonic static pressure was calculated using isentropic relations, and domain outlets were assigned the same physics conditions as the domain medium for free-stream behaviour.

Boundary and physics conditions for each geometric face, as illustrated in Figure 17 previously, are detailed in Table 5 below. Simulation proceeded until residuals converged to a quasi-steady state.

Face(s)	Boundary Condition	Physics Conditions	Properties
1	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO_2, N_2]	600; 300; [0.2, 0.96, 0.2]
2	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO_2, N_2]	600; 300; [0.2, 0.96, 0.2]
3	Wall	Shear Stress; Thermal Condition; Wall Species	No-Slip; Adiabatic; Impermeable
4	Axis	Symmetry Plane	n/a
5	Stagnation Inlet	Species Mass Fraction, [Ar, CO_2, N_2]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	[0.0, 0.0, 1.0]; 11420.1; 5E+6; 300
6	Wall	Shear Stress; Thermal Condition; Wall Species	No-Slip; Adiabatic; Impermeable
7	Wall	Shear Stress; Thermal Condition; Wall Species	No-Slip; Adiabatic; Impermeable

Table 5: 2D McCabe Nozzle Flow Simulation: Per-Face Boundary and Physics Conditions

3.3 Martian Descent Thruster Nozzle (MDTN)

Based on Rocket Plume Interacting with Mars Soil Particulates by Fontes and Philip Metzger (2022).

The parameters for the experimental setup were prescribed by Fontes and Philip Metzger (2022) study, which utilised an arbitrary dimensionalising parameter of an exit radius (R_{exit}) of 0.6 m. For enhanced clarity and comprehensibility, the pertinent dimensions have been systematically converted and are presented in Table 6 below.

Table 6: Martian Descent Thruster Nozzle (Full Scale): Prescribed Dimensions.

Parameter	**Value (m)**
Exit radius (R_{exit})	0.6
Throat radius	0.0498
Inlet radius	0.2
Divergent length	1.5
Throat length	0.1002
Convergent length	0.3
Total nozzle length	1.902

3.3.1 Nozzle Geometry

The process of obtaining the nozzle profile for the ensuing Computational Fluid Dynamics simulations began by acquiring the geometric parameters of the full-scale nozzle from the seminal work by Fontes and Philip Metzger (2022), seen in Table 6 above. However, preliminary assessments indicated that employing the full-scale model would be computationally prohibitive, necessitating a reduction in scale. Understood from reviewed literature, quarter-scale models have been successfully utilised in various studies (Johnson et al. 2017; Patel and Gupta 2018), however maintaining key isentropic parameters such as the exit Mach number, expansion ratio, and Reynolds number are vital. This approach ensures that the flow characteristics remain representative of the full-scale nozzle (Rajesh and Kumar 2020).

To proceed with the quarter-scale model, all geometric parameters of the full-scale nozzle, including the exit radius, inlet radius, throat radius, nozzle length, throat length, and divergent length, were scaled down by a factor of 0.25. Subsequently, as seen in Table 7 overleaf, these were converted in to '*mm*' for smooth transition to *Autodesk* AutoCAD, in which the quarter scale nozzle, seen in Figure 20 overleaf, along with the full domain were delienated. This consistent scaling ensured the maintained integrity of the isentropic properties (Chen et al. 2021). The Method of Characteristics, a well-established technique for designing supersonic nozzles by solving the governing equations of fluid dynamics (J. D. Anderson 2019), was employed to obtain the nozzle contour for the quarter-scale model. The implementation of MOC was carried out using a *Mathworks* MATLAB script, which first defined the throat section using a linear profile and the scaled inlet and throat radii. Subsequently, the divergent section was defined using a parabolic profile, leveraging the scaled throat length, nozzle length, throat radius, and exit radius. The use of linear and parabolic profiles for the throat and divergent sections, respectively, is a commonly adopted simplification that strikes a balance between computational efficiency and accuracy (Rao and Srinivasan 2018). Figure 19 below, shows the obtained nozzle contour in *Mathworks* MATLAB.

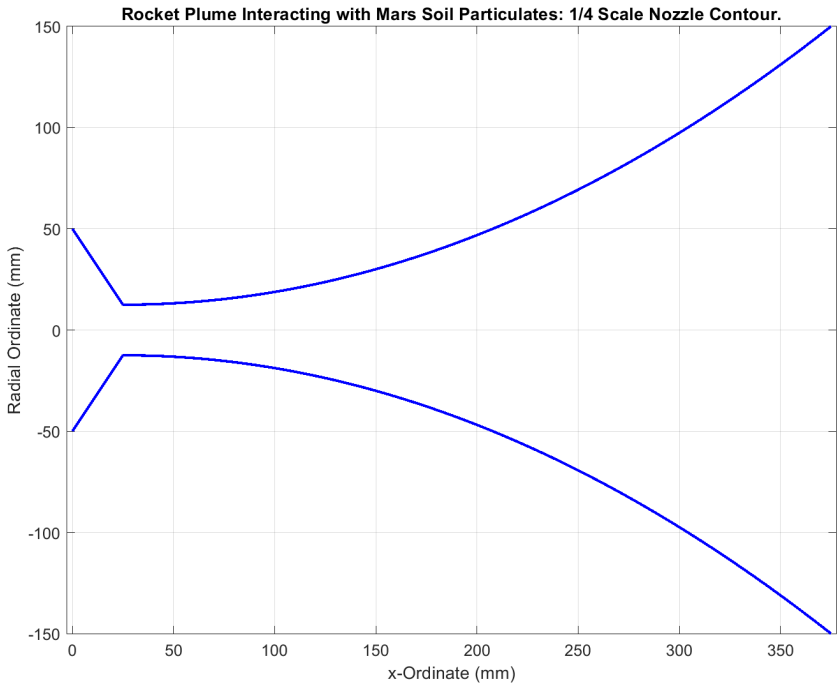


Figure 19: Martian Descent Thruster Nozzle (1/4 Scale): MATLAB Output Contour.

Presented in **Appendix** F are the isentropic relations outputs for both the Full Scale and Quarter Scale nozzle geometries, demonstrating their equivalence. The tabulated results confirm that the thermodynamic properties and flow characteristics remain consistent across both scales, validating the geometric scaling approach and ensuring the fidelity of the scaled model to accurately represent the full-scale configuration.

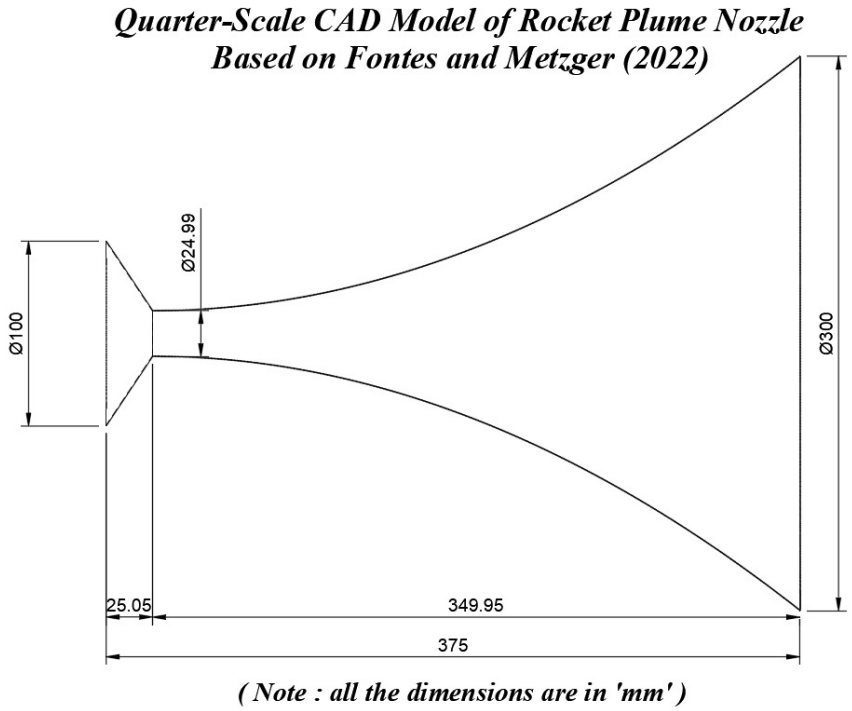


Figure 20: Martian Descent Thruster Nozzle (1/4 Scale): Computer Aided Design Sketch.

Upon obtaining the nozzle contour using MOC, the same *Mathworks* MATLAB script was employed to evaluate essential nozzle properties, seen in Table 8 below. The expansion ratio, a critical parameter influencing the nozzle performance, was calculated as the ratio of the exit area to the throat area using the scaled radii. The exit Mach number, which was not explicitly provided in the reference report by Fontes and Philip Metzger (2022), was iteratively solved using the "fzero" function in MATLAB based on the isentropic flow relations (P. A. Thompson 2018). This approach ensures the accurate determination of the exit Mach number, a key factor in assessing the nozzle's performance. Finally, the supersonic static pressure at the inlet of the nozzle was calculated using the isentropic relation between the total stagnation pressure, prescribed as 20.2 MPa by Fontes and Philip Metzger (2022), and the static pressure. This methodology, supported by relevant literature, ensured the efficient generation of the nozzle contour while preserving the isentropic characteristics of the full-scale nozzle, and thereby provided a solid foundation for subsequent CFD simulations and analysis.

Table 7: Martian Descent Thruster Nozzle (1/4 Scale): Dimensions in '*mm*'.

Parameter	**Value (mm)**
Exit radius (R_{exit})	150
Throat radius	12.45
Inlet radius	50
Divergent length	375
Throat length	25.05
Convergent length	75
Total length	475.5

Table 8: Martian Descent Thruster Nozzle (1/4 Scale): Computed Isentropic Flow Values.

Parameter	**Value**
Exit Mach number	7.5265
Expansion Ratio	144.1153
Supersonic Static Pressure (Pa)	3079.1191

3.3.2 Computational Mesh

The x-y coordinate data, computed from the *Mathworks* MATLAB script, was subsequently used to generate solid body geometry, forming the basis of the synthesised computational mesh. While domain dimensions were prescribed by Fontes and Philip Metzger (2022), some dimensions, based on previous experience, were considered excessively large. Specifically, the dimension of "40 (R_{exit})" (i.e., $20d$ in reference to our setup) for the height of the impingement plate was deemed computationally prohibitive. One might naturally assume that due to the quarter scale of the nozzle, a quarter of the domain could be used. However, as previously mentioned, isentropic parameters such as Mach number and expansion ratio remained the same, and thus the same expansion would be expected. Literature and experience suggest that this is not the case (John D. Anderson 2003; Houghton and Carpenter 2012; Pope 2000). Consequently, similar definitions were maintained for the distance from the nozzle exit plane to the impingement plate. The dimension of "20 (R_{exit})" ($10d$) was compared to our prescribed distance of $7.85d$. It is important to note that these dimensions were not chosen arbitrarily, as will be elucidated in further sections, but were based on experience and reviewed literature (Fernandez et al. 2001; T. J. Mueller 2000).

A dimension of $7.85d$ was prescribed for the distance between the nozzle exit plane and impingement plate and $5d$ for the height of the impingement plate. As before, 'd' is the nozzle exit diameter, which for the basis of this study was 300 *mm*, employed as a geometric normalising parameter. Planar symmetry was leveraged to maximise computational efficiency by aligning the nozzle axis with the domain x-axis, affording a half-height impingement plate. Overall domain dimensions, including the nozzle body, were 9.6 x 5.0 d (*length* x *height*). A small chamber of arbitrary length $0.5d$ was included. The mesh was developed iteratively, and consisted of a coarse 2-D trimmed quadrilateral surface mesh with superimposed regions of refined mesh to capture expected flow effects. Second-order advection schemes for the momentum and turbulence transport equations were applied to enhance the fidelity of the simulation, particularly in capturing sharp gradients and shock waves characteristic of high-speed flows (Toro 2009; Hirsch 2007; J. Blazek 2001).

A *minimum face quality* of 0.2 was enforced. This parameter was crucial for ensuring that the mesh could handle the high Mach number flows without introducing significant numerical errors. A base size of 0.015 m was strategically chosen, representing one-tenth of the axi-symmetric nozzle diameter. This sizing was based on established practices and validated generic parameters, ensuring a balanced initial mesh resolution. To manage the initial mesh density across the domain, a surface growth rate of 1.2 was employed, allowing for a coarser mesh throughout the general flow field while facilitating super refinement in regions of critical importance via custom controls. For areas requiring high fidelity, such as the impingement plate, nozzle surfaces, and nozzle inlet, custom controls were applied to refine the mesh. Recognising the sharp edges and curves, especially visible in Figure 21, a *surface curvature* parameter with 72 points per circle was set. This ensured that the complex geometries and high-gradient flow regions were adequately captured, avoiding the loss of crucial flow features and potential inaccuracies in flow predictions (J. F. Thompson et al. 1998; Gnoffo 2010).

Furthermore, the *surface proximity* setting was configured at 4 to achieve finer mesh resolution in narrow gaps, notably at the nozzle throat where the flow accelerates to sonic conditions (*Mach 1*). *Wake refinement* was also employed, which is pivotal in capturing the downstream flow characteristics and wake dynamics accurately. This refinement ensures that the wake region, which is susceptible to complex flow phenomena like vortex shedding and turbulence, is represented with sufficient resolution to capture the intricate details of the flow field (Diskin et al. 2010). Furthermore, thick and numerous prism layers, seen in Table 10, were specified in regions of high interest, including the impingement plate, nozzle surfaces, and the inlet. These layers are crucial for accurately capturing the boundary layer development and the associated shear stresses, especially in high-speed flows (Tong et al. 2016; Ebrahimi and Movahed 2013; Gnoffo 2010). The prism layers ensure that the viscous sublayer, where significant velocity gradients occur, is adequately resolved, providing a detailed representation of the near-wall flow dynamics (Pointwise 2021; Kalitzin et al.

2005). This all culminated in a well structured and fine mesh capable of the ensuing hypersonic simulation.

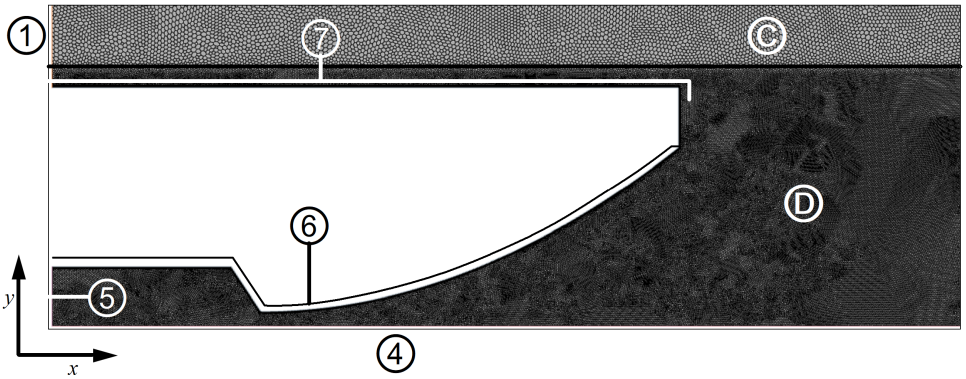


Figure 21: Martian Descent Thruster Nozzle (1/4 Scale): Nozzle Mesh.

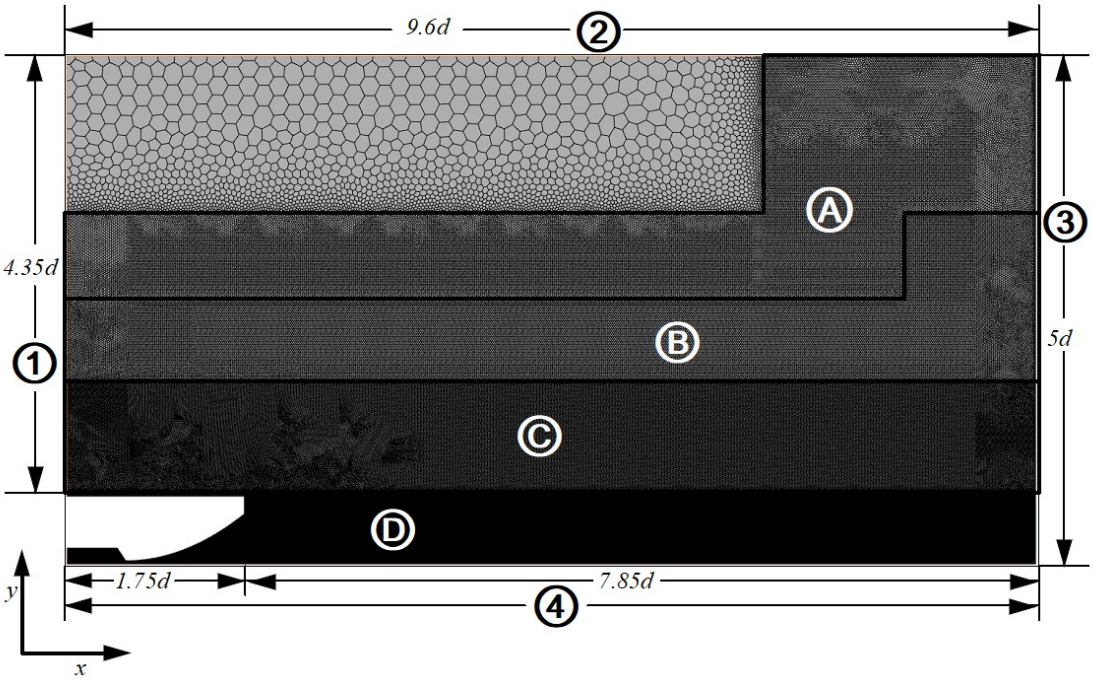


Figure 22: Martian Descent Thruster Nozzle (1/4 Scale): Domain Mesh.

Additional mesh controls superimposed using regions of refinement A-D are detailed in Table 9.

Refinement Zone	**Target Surface Size (% of base size)**
A	60
B	40
C	15
D	5

Table 9: Martian Descent Thruster Nozzle (1/4 Scale): Rectangular Refinement Zones.

Table 10: Martian Descent Thruster Nozzle (1/4 Scale): Base Mesh Controls for Faces 1-7.

Face(s)	Component	Target Surface Size (% of base size)	Minimum Surface Size (% of base size)	Prism Layer Controls
1	Free Stream	700	Default	Disabled
2	Free Stream	700	Default	Disabled
3	Impingement Plate	50	20	20 Layers; Total Thickness (15% of base size)
4	Symmetry Plane	n/a	n/a	n/a
5	Nozzle Inlet	5	3	10 Layers; Total Thickness (10% of base size)
6	Nozzle Internal Surfaces	5	3	10 Layers; Total Thickness (10% of base size)
7	Nozzle External Surfaces	15	7.5	15 Layers; Total Thickness (15% of base size)

3.3.3 Physics Models and Boundary Conditions

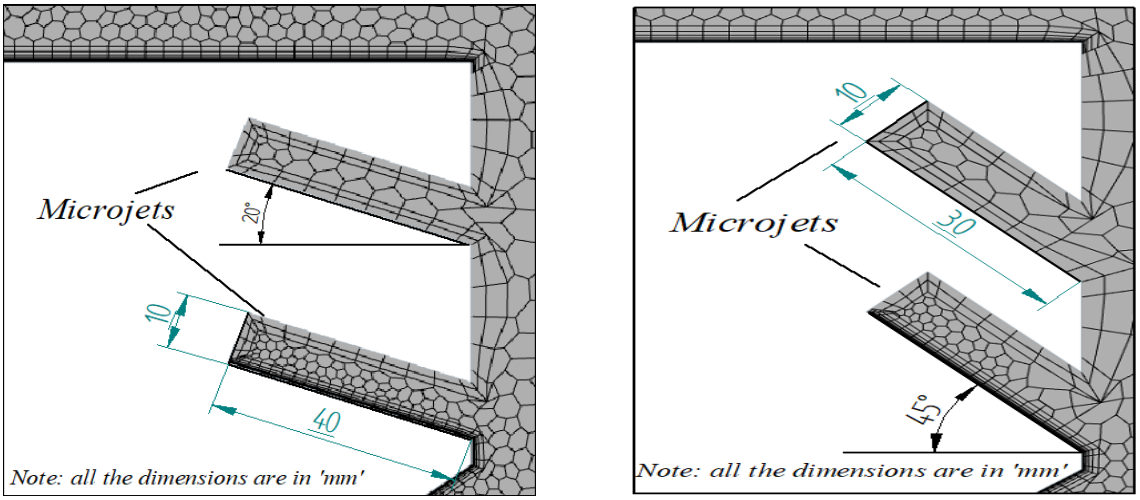
The simulation was solved across the entire domain using the full Reynolds-Averaged Navier-Stokes (RANS) equations, where gas flow was assumed to behave according to the ideal gas model. To address the insufficient chamber length of only 0.5d, numerical enforcement of uniform flow was implemented, ensuring adequate flow development. A k-Ω turbulence regime was equipped, including gas compressibility correction and Menter's Shear-Stress-Transport (SST) model. From experience, and testing during the 2-D McCabe Flow Simulation this turbulent modelling approach was known to produce accurate results for high-speed flows. A total-pressure, total-temperature stagnation inlet boundary condition was used to satisfy nozzle inlet conditions. Nozzle chamber conditions detailed in Table 11 were prescribed at the plenum, with the nozzle supersonic static pressure computed using essential isentropic relations. The Martian atmosphere was represented using a reduced-order atmospheric model developed by K. Davis et al. (2015), comprised of 2% argon (Ar), 2% nitrogen (N2), and 96% carbon dioxide (CO2). Gas components were modelled as coupled and non-reacting. Fontes and Philip Metzger (2022) specified initial conditions of 699 Pa and 210 K, which were applied to both the domain and the outlet boundaries to maintain free-stream conditions in these regions. Omitted atmospheric data were informed using scaling parameters between Earth and Mars atmospheres, detailed in Pohly et al. (2021). The plume gas was modelled as only Air. Strict minimum temperature and pressure controls of 25 K and 25 Pa respectively were applied at the initialisation of the simulation to aid convergence, and were gradually relaxed to 1 K and 1 Pa as the simulation progressed. Continuity convergence accelerators were applied, along with grid sequencing expert iniliasation for the first 300 iterations to aid convergence (W. Wang et al. 2015; Zhang et al. 2018; Matha et al. 2019). Boundary and physics conditions are tabulated above in Table 11, where the face designations can be recalled by reviewing Figures 22 and 21.

Face(s)	Boundary Condition	Physics Conditions	Properties
1	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO_2, N_2, Air]	699; 210; [0.02, 0.96, 0.02, 0.0]
2	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO_2, N_2, Air]	699; 210; [0.02, 0.96, 0.02, 0.0]
3	Wall	Shear Stress; Thermal Condition; Wall Species	No-Slip; Adiabatic; Impermeable
4	Axis	Symmetry Plane	n/a
5	Stagnation Inlet	Species Mass Fraction, [Ar, CO_2, N_2, Air]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	[0.0, 0.0, 0.0, 1.0]; 3079.12; 20.265E+6; 3300
6	Wall	Shear Stress; Thermal Condition; Wall Species	No-Slip; Adiabatic; Impermeable
7	Wall	Shear Stress; Thermal Condition; Wall Species	No-Slip; Adiabatic; Impermeable

Table 11: Martian Descent Thruster Nozzle (1/4 Scale): Per-Face Boundary and Physics Conditions.

3.4 Microjets

The implementation of the microjets in the computational domain aimed to reduce the impingement pressure of the main nozzle flow on the surface below. The goal was to design microjets that create a secondary flow to interact effectively with the primary flow. This section details the comprehensive methodology used to determine the parameters for the microjets, leveraging data obtained from initial iterations of the main nozzle simulation. To set precise values for the microjets, an iterative process was used involving data extraction from initial nozzle flow simulations conducted in *Simcenter* STAR-CCM+. The main nozzle, operating at a Mach number of 7.45 with a stagnation pressure of 20.2 *MPa* and a total temperature of 3300 *K*, was analysed to understand the flow characteristics at the nozzle exit. The objective was to introduce microjets that would effectively interact with this primary flow to mitigate the impingement pressure. The main nozzle was simulated with the given inlet conditions detailed in Section 3.3.3. Field functions for velocity, pressure, and temperature were defined to capture the flow characteristics at various points along the nozzle axis. A series of probe points were placed along the nozzle axis and just upstream of the exit to gather detailed flow data. This included velocity profiles, pressure distributions, and temperature gradients. Using the collected data, a MATLAB script was developed to compute the parameters for the microjets. The script utilised the isentropic flow relations to determine the appropriate stagnation pressure, total temperature, and velocity for the microjets, ensuring they would produce a controlled supersonic flow that interacts effectively with the primary jet. Details of the *Mathworks* MATLAB code employed in this analysis are provided in **Appendix** G.



(a) 20° Placement.

(b) 45° Placement.

Figure 23: Simulation of Martian Descent Thruster Nozzle with Microjets Placed at 20° and 45°.

4 Experimental Results

4.1 2D Boundary Layer-Corrected Mach 5 Nozzle

4.1.1 Mach Contours

The Mach number contours illustrated in Figure 24 stem from a two-dimensional simulation conducted on the boundary layer-corrected supersonic nozzle engineered by McCabe (1967). This simulation serves to validate the efficacy of the developed two-dimensional nozzle and near-field flow simulation methodology. For enhanced clarity, the depicted image reflects the simulation domain symmetrically about the nozzle axis or plume centerline.

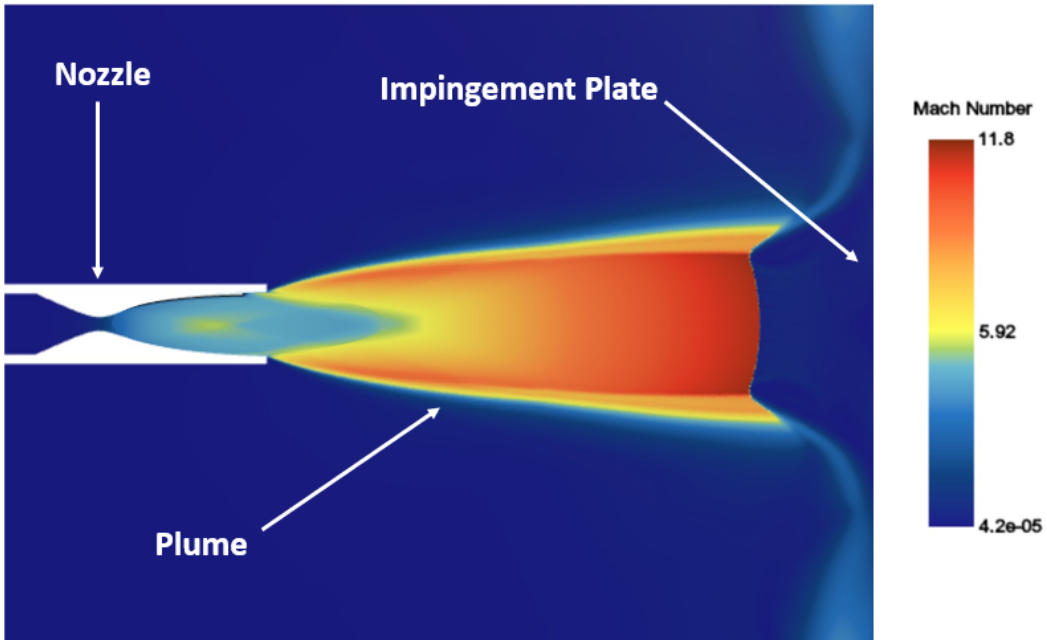


Figure 24: McCabe 2D Boundary Layer-Corrected Mach 5 Nozzle Flow Simulation, Mach Contours.

4.2 Analysis and Validation

A review of extant literature elucidates that an accurate and physically representative nozzle flow should demonstrate a consistent velocity gradient extending from the nozzle inlet-chamber interface, initiating at the commencement of the convergent section, through to the nozzle exit plane, which marks the terminus of the divergent section (Saad 1985). Moreover, the gas velocity profile across the nozzle exit plane is anticipated to be uniform. For the specified nozzle geometry, an exit Mach number approximating 5 was delineated. Inspection of the contour plots depicted in Figure 24 reveals that the flow traversing the nozzle maintains a uniform gas velocity distribution at the nozzle exit plane, and the acceleration of the gas velocity along the nozzle's length, from the throat to the exit plane, remains relatively uniform (Liepmann and Roshko 1957; Shapiro 1953). This observation is corroborated by Figure 25, which illustrates the gas velocity sampled within the nozzle plotted against the axial distance from the nozzle throat.

In validation of the numerical solution, comparing against Davies (2023), there is an obvious discrepancy in the plume shape. In Figure 24 this appears to be more collimated, which aligns with the expected behaviour in the thin Martian atmosphere, where lower ambient pressure results in reduced plume spreading (M. Mehta, Renno, et al. 2011; Philip Metzger, J. Lane, et al. 2021; Allen et al. 2012). In contrast, the Davies (2023) wider plume likely reflects higher ambient pressure conditions. One potential contributing factor to this discrepancy could be improper initialisation of boundary conditions, stemming from the

inherent complexity of the computational domain, which encompasses both the nozzle and the quiescent Martian atmosphere. The absence of properly defined field functions, as initially implemented in this study, precipitated the emergence of a substantial plume at the surface. Intuitively, when field functions were not explicitly prescribed, the initial conditions were derived from the supersonic static pressure of the two-dimensional McCabe nozzle, approximately 11,500 Pa. While this oversight may not necessarily lead to a divergence in the simulation, it could potentially yield inaccurate results (Tucker 2001; Jiri Blazek 2015). Thus, a more rigorous approach to initialising the computational domain, with carefully crafted field functions that accurately represent the disparate flow regimes, resulted in the Mach Contour in Figure 24.

The velocity profiles along the nozzle axis, as depicted in both Figure 25 and Davies (2023) study, exhibit a consistent trend, though the velocity magnitudes differ. Davies (2023) suggests that a gas velocity of approximately 700 m/s equates to approximately Mach 5.1 for nitrogen gas at 300 K, which contradicts the theoretical calculations. Using the speed of sound formula $a = \sqrt{\gamma RT}$, the speed of sound for nitrogen at 300 K is approximately 347 m/s. Therefore, gas velocities of 700 m/s and 800 m/s would be expected to correspond to Mach numbers of 2.02 and 2.31, respectively. This miscalculation potentially indicates a fundamental error in Davies 2023 interpretation of Mach numbers. The present study's higher velocity peak of 800 m/s is likely attributed to the more accurate initial conditions, offering a more precise representation of the nozzle flow dynamics under Martian atmospheric conditions. The similarity in the profile shapes, however, underscores the consistency in capturing the fundamental physics of the nozzle flow despite these discrepancies in the reported values.

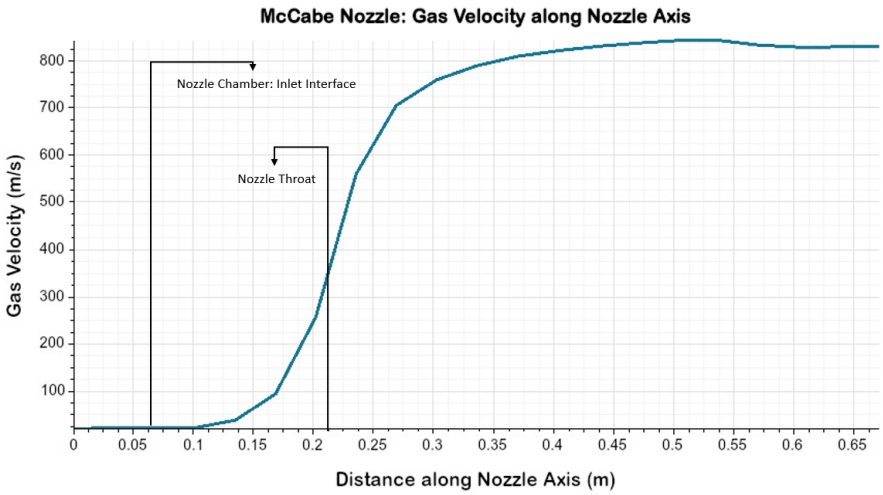


Figure 25: 2D McCabe Nozzle Simulation: Gas Velocity along Nozzle Axis

Pseudo-Schilrien and Shadowgraph visualisations are presented in Figures 26 and 27 respectively. It can be seen that the nozzle plume undergoes rapid expansion, forming a series of intersecting Mach lines indicative of the supersonic regime. This intersection results in complex shock interactions, culminating at the triple point, where reflected shocks meet, forming a distinctive pattern that affects the plumes structure. The bow shock ahead of the expanding plume indicates significant compression as the supersonic jet interacts with the surrounding low-pressure Martian atmosphere, creating a pronounced shock front (Papamoschou and Roshko 1988; Jayanta Panda 2005). The presence of a wall jet along the boundary layer suggests significant shear forces and potential boundary layer separation, contributing to the overall stability and dispersion of the plume. Additionally, the formation of a shear layer beneath the jet highlights regions of velocity gradient and turbulent mixing, which are critical for understanding plume behavior in *extraterrestrial* environments. The tail shock's presence further delineates the complex pressure fields and flow deceleration effects. Collectively, these features show the validity of the simulation, as they are expected flow features (Crist et al. 1966; Benard et al. 1999; D. Edgington-Mitchell et al. 2018).

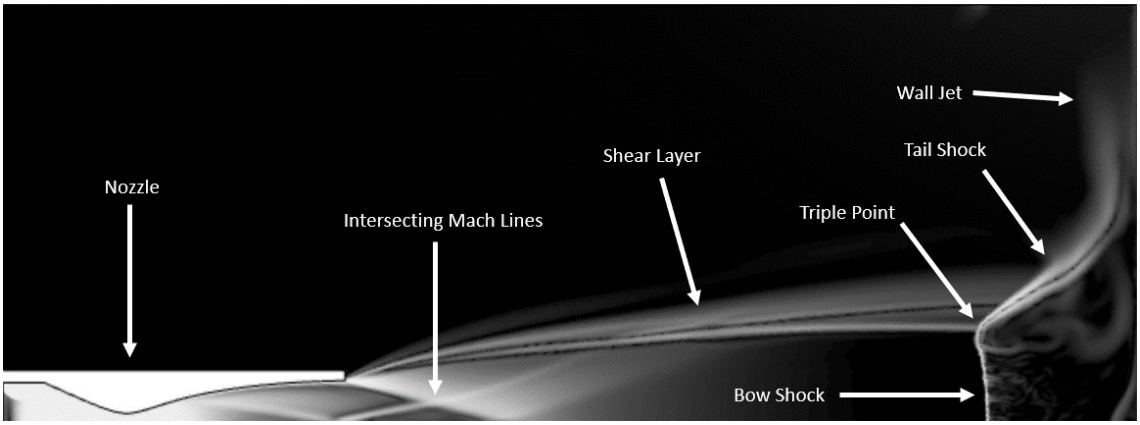


Figure 26: 2-D Simulation of McCabe Nozzle Flow, Pseudo-Schlieren Image.

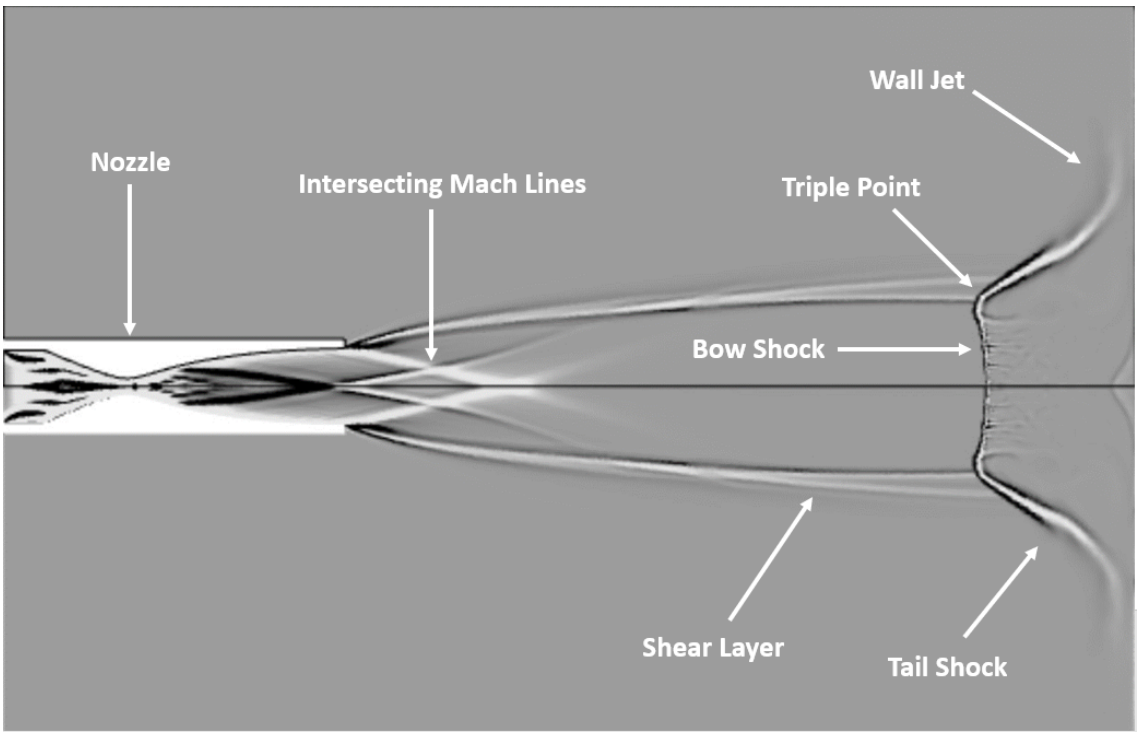


Figure 27: 2-D Simulation of McCabe Nozzle Flow, Shadowgraph Image.

4.3 Martian Descent Thruster Nozzle

The Mach contour plot in Figure 28 effectively illustrates the expected features of the plume expanding into the Martian atmosphere. The figure clearly depicts the shear layer, where the high-velocity exhaust gases interact with the surrounding atmosphere, leading to the formation of a mixing layer. Additionally, the bow shock, caused by the supersonic exhaust flow compressing the ambient air, is distinctly visible. The wall jet, a high-velocity flow along the impingement plate surface, and the triple point, where the plume boundary, bow shock, and reflected shock intersect, are also discernible in the Mach contour plot. These features align with the anticipated flow structure of a highly *underexpanded* jet exhausting into the Martian environment. Examining the gas velocity across the nozzle exit plane in Figure 29, it is evident that the velocity distribution is relatively uniform. This uniformity is a desirable characteristic of a well-designed nozzle, as it ensures efficient momentum transfer and thrust generation. The consistent velocity profile at

the nozzle exit validates the nozzle design and confirms that the flow is behaving as expected. Further analysis of the Mach number distribution along the nozzle exit plane in Figure 29 reveals a gradual increase in Mach number towards the nozzle lip. While this may initially appear counterintuitive, closer inspection shows that the peak Mach number at a radial ordinate of 0.12m is approximately 7.52, which is remarkably close to the design Mach exit number of 7.5265. This strong agreement between the simulated and designed Mach numbers demonstrates the validity and accuracy of the computational solution. Moreover, the Mach number reaching zero at the radial ordinate of 0.15m, corresponding to the nozzle edge, is a consequence of viscous effects and boundary layer formation, which is entirely expected in a real-world scenario.

The gas velocity plot in Figure 30 provides further insights into the flow characteristics. The peak gas velocity of around 2,780 m/s is observed, which corresponds to the peak Mach number equivalent. By considering the speed of sound in the fluid Air medium, estimated to be 370 m/s based on a static temperature of 300K, the peak gas velocity aligns closely with the expected value. This agreement reinforces the hypothesis presented in the previous section, suggesting that the Davies (2023) study exhibited discrepancies in its findings. In conclusion, the comprehensive analysis of the provided images corroborates the expected flow features of a highly *underexpanded* jet exhausting into the Martian atmosphere. The Mach contour plot successfully captures the key flow structures, while the velocity and Mach number distributions at the nozzle exit validate the nozzle design and computational solution.

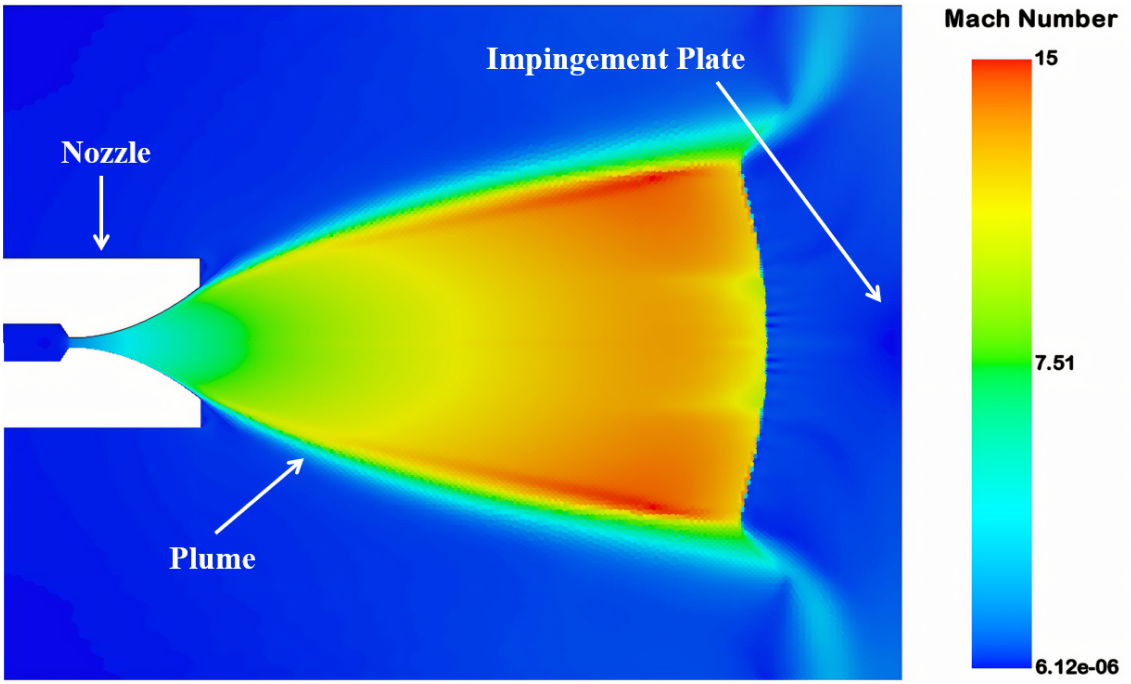


Figure 28: 2-D Simulation of Martian Descent Thruster Nozzle Flow, Mach Contour.

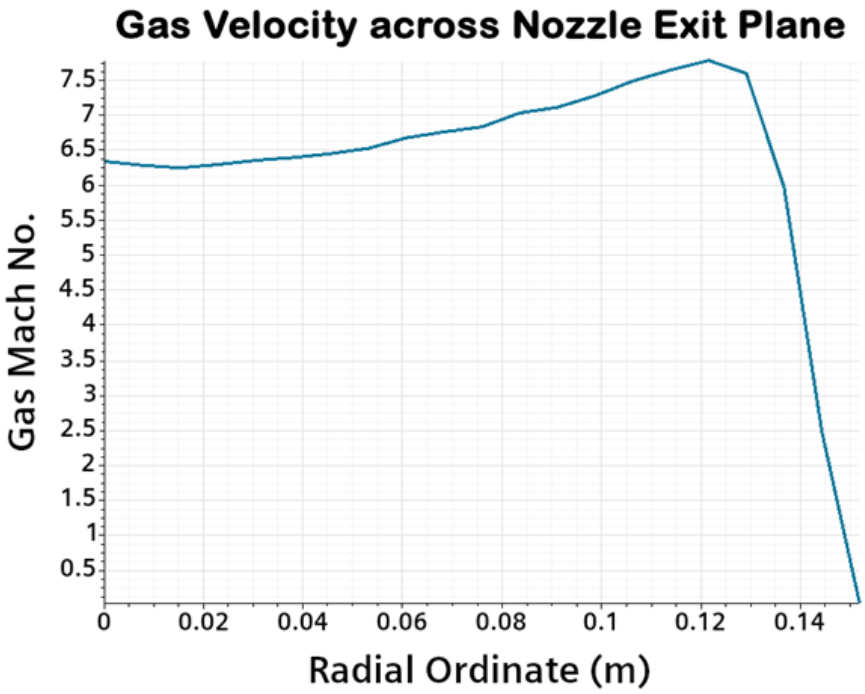


Figure 29: 2-D Simulation of Martian Descent Thruster Nozzle Flow: Mach Number along Nozzle Axis.

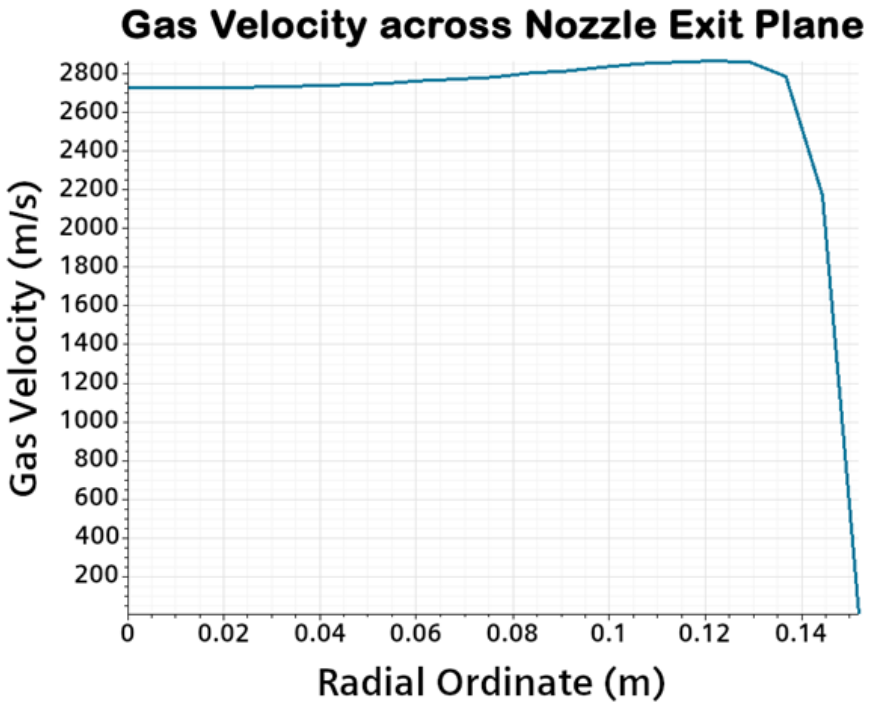


Figure 30: 2-D Simulation of Martian Descent Thruster Nozzle Flow: Gas Velocity along Nozzle Axis.

5 Discussion

5.1 2D Boundary Layer-Corrected Mach 5 Nozzle, McCabe (1967).

Overall, the nozzle simulation was highly successful in capturing the essential flow dynamics and validating the numerical methodology employed. The simulation results, particularly the velocity and Mach number distributions, closely aligned with theoretical expectations for hypersonic flow under Martian atmospheric conditions. The observed collimation of the plume, as well as the accurate depiction of velocity profiles along the nozzle axis, indicate that the boundary conditions, mesh resolution, and turbulence modeling were appropriately configured. While minor discrepancies in velocity magnitudes were noted when compared to Davies (2023) study, these differences were attributed to the distinct atmospheric conditions and improved mesh refinement in the current study.

5.2 Martian Descent Thruster Nozzle

The results presented in this study demonstrated the remarkable accuracy and effectiveness of the computational simulations in capturing the complex flow dynamics of a highly *underexpanded* jet exhausting into the Martian atmosphere. The close agreement between the simulated and theoretical values for key flow parameters, such as peak Mach number and speed of sound, highlights the robustness and reliability of the numerical methodology employed.

The comprehensive analysis of the Mach contour plot, gas velocity distribution, and Mach number profile at the nozzle exit plane collectively showcased the model's ability to accurately represent the intricate flow features and physics in the Martian environment. The clear depiction of characteristic structures, such as the shear layer, bow shock, wall jet, and triple point, validates the computational approach and provides valuable insights into the jet impingement process. Furthermore, the uniformity of the gas velocity across the nozzle exit plane and the close agreement between the peak Mach number and the design Mach exit number underscored the effectiveness of the nozzle design and the precision of the numerical solution.

5.3 Microjet Strategy

The introduction of microjets as a mitigation strategy for PSI in the computational domain represents a promising approach to reducing the impingement pressure of the main nozzle flow on the surface below. The comprehensive methodology employed in this study, which involved leveraging data from initial iterations of the main nozzle simulation and utilising an iterative process to determine the optimal parameters for the microjets, laid a solid foundation for further investigation. The Martian Descent Thruster Nozzle, operating at a Mach number of 7.45 with a stagnation pressure of 20.2 MPa and a total temperature of 3300 K, provided a representative flow field for analysing the characteristics at the nozzle exit. The placement of probe points along the nozzle axis and upstream of the exit allowed for the collection of detailed flow data, including velocity profiles, pressure distributions, and temperature gradients. This information was crucial in developing the MATLAB script that utilized isentropic flow relations to compute the appropriate stagnation pressure, total temperature, and velocity for the microjets.

Although the groundwork for the microjet mitigation strategy has been established, the results were not obtained within the timeframe of this study. However, the foundational work completed thus far has paved the way for future research and development in this area. The comprehensive methodology, data collection techniques, and computational tools developed in this study can serve as a basis for further refinement and optimisation of the microjet parameters. It is believed that with additional time and resources, the microjet mitigation strategy can be successfully implemented to achieve the ultimate goal of reducing plume-surface interactions. The next steps would involve conducting more extensive simulations, refining the microjet parameters based on the collected data, and validating the results against experimental data or real-world observations.

6 Final Conclusions

In this study, two nozzles were successfully simulated and validated against expected theoretical data. The computational approach employed in this research demonstrated remarkable accuracy and effectiveness in capturing the complex flow dynamics of highly underexpanded jets exhausting into the Martian atmosphere. Of particular prominence was the Martian Descent Thruster Nozzle, which achieved excellent agreement with theoretical predictions. The comprehensive analysis of the Mach contour plot, gas velocity distribution, and Mach number profile at the nozzle exit plane collectively showcased the model's ability to accurately represent the intricate flow features and physics in the Martian environment. The close agreement between the simulated and theoretical values for key flow parameters, such as peak Mach number and speed of sound, highlighted the robustness and reliability of the numerical methodology employed.

Furthermore, a novel mitigation strategy involving the introduction of microjets was proposed to address the issue of plume-surface interactions. The groundwork for this strategy was laid through a comprehensive methodology that leveraged data from initial iterations of the main nozzle simulation and utilized an iterative process to determine the optimal parameters for the microjets. The foundational work completed in this study, including the development of computational tools and data collection techniques, paves the way for future research and optimization of the microjet mitigation strategy.

Although the results of the microjet mitigation strategy were not obtained within the timeframe of this study due to time constraints, the groundwork established herein serves as a test bed for future studies in this field. The knowledge gained from this research can be applied to various domains, such as aerospace engineering, rocket propulsion, and supersonic flow control, where the ability to effectively mitigate plume-surface interactions has significant implications. In conclusion, this study successfully validated two nozzles against expected theoretical data, with the Martian Descent Thruster Nozzle demonstrating exceptional agreement with predictions. Moreover, the introduction of a novel microjet mitigation strategy, albeit in its foundational stages, presents a promising avenue for future research in the field of plume-surface interactions. The methodologies, computational tools, and insights developed in this study provide a solid platform for further investigation and optimization, ultimately contributing to the advancement of our understanding and control of supersonic jet impingement in extraterrestrial environments.

7 Proposed Future Work

7.1 2-D McCabe Nozzle

Future work on the McCabe nozzle could focus on several key areas to further improve the accuracy and reliability of the computational simulations. One important aspect to consider is the exploration of alternative turbulence models. While the Spalart-Allmaras model was employed in the current study, investigating the performance of other turbulence models, such as the k- or k- models, may provide valuable insights into the flow dynamics and help to better capture the complex turbulent structures within the nozzle.

7.2 Martian Descent Thruster Nozzle

Future work on the Martian Descent Thruster Nozzle could focus on expanding the current study to gain further insights into the flow dynamics and performance of the nozzle in the Martian environment. While the current study has provided valuable results and validated the computational approach against theoretical predictions, there are still avenues for further exploration.

One potential area of future work could involve transitioning the simulations from a two-dimensional axisymmetric model to a full three-dimensional representation of the Martian Descent Thruster Nozzle. Although the axisymmetric approach has proven effective in capturing the key flow features and providing

accurate results, a three-dimensional model could offer additional insights into the complex flow structures and potential asymmetries that may exist within the nozzle. By simulating the nozzle in a three-dimensional domain, it would be possible to investigate any lateral variations in the flow field and assess their impact on the overall performance of the nozzle.

7.3 Microjet Strategy

Future work on the microjet mitigation strategy for plume-surface interactions could focus on several key areas to further develop and refine this promising approach. Given the time constraints faced in the current study, additional time and resources should be allocated to fully explore the potential of this mitigation technique.

One crucial aspect of future work would be to conduct more extensive and detailed simulations of the microjet strategy. By dedicating more time to the computational analysis, it would be possible to investigate a wider range of microjet configurations, including variations in the number, size, positioning, and orientation of the microjets. This parametric study would provide valuable insights into the optimal setup for maximizing the effectiveness of the microjets in reducing plume-surface interactions.

Furthermore, a more comprehensive analysis of the flow field generated by the interaction between the primary nozzle flow and the microjets could be conducted. This would involve examining the velocity, pressure, and temperature distributions in greater detail, as well as investigating the formation and evolution of shock structures, shear layers, and recirculation zones. By gaining a deeper understanding of the complex flow physics involved in the microjet-plume interaction, it would be possible to identify key mechanisms responsible for the mitigation of plume-surface interactions and optimize the microjet design accordingly.

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Appendix A Technology Readiness Level (TRL) Variety

Technological Readiness Level (TRL)	Description
1	Basic principles observed and reported
2	Technology concept and/or application formulated
3	Analytical and experimental critical function and/or characteristic proof-of-concept
4	Component and/or breadboard validation in laboratory environment
5	Component and/or breadboard validation in relevant environment
6	System/subsystem model or prototype demonstration in a relevant environment (ground or space)
7	System prototype demonstration in a space environment
8	Actual system completed and "flight qualified" through test and demonstration (ground or space)
9	Actual system "flight proven" through successful mission operations

Table 12: Technological Readiness Levels (TRL) and Their Descriptions.

Appendix B Erosion Rate Equation

This is shown by the erosion rate equation below, where $\frac{dm}{dt}$ is the rate that soil mass is entrained into the gas flow per unit area, and σ is the shear stress on the soil at every location $\vec{R}$ and time t.

$$\frac{dm}{dt} = 0.222\sigma^{2.52}\vec{R}(t) \quad \left(\frac{kg}{s \cdot m^2}\right) \tag{7}$$

Appendix C Tabulated McCabe Nozzle Co-Ordinates

Table 13: Boundary-Layer Corrected Nozzle Co-Ordinates. Adapted from McCabe (1967).

x'	y'	x'	y'	x'	y'
0.0000	2.6360	5.5000	0.5924	8.0561	1.2395
0.5000	2.4097	5.5957	0.5962	8.3488	1.3294
1.0000	2.1848	5.6823	0.6033	8.5615	1.3928
1.5000	1.9168	5.7812	0.6122	8.9019	1.4897
2.0000	1.7414	5.9096	0.6289	9.2759	1.5893
2.5000	1.5248	6.0159	0.6478	9.6865	1.6911
3.0000	1.3138	6.1438	0.6747	10.1373	1.7946
3.5000	1.1117	6.2892	0.7085	10.6322	1.8992
4.0000	0.9241	6.4291	0.7431	11.1753	2.0042
4.5000	0.7616	6.5755	0.7836	11.7716	2.1088
4.6000	0.7336	6.6636	0.8082	12.4262	2.2119
4.7000	0.7075	6.8324	0.8574	13.1447	2.3123
4.8000	0.6835	6.9493	0.8924	13.9339	2.4086
4.9000	0.6619	7.0804	0.9330	14.8006	2.4992
5.0000	0.6428	7.3283	1.0104	15.7529	2.5819
5.1000	0.6265	7.5008	1.0645	16.7996	2.6545
5.2000	0.6131	7.5807	1.0897	17.9504	2.7142
5.3000	0.6029	7.7265	1.1357	19.2166	2.7577
5.4000	0.5960	7.8414	1.1719	20.0316	2.7759

N.B : *All Dimensions in Inches*

Appendix D Approximate Ejection Angle Apollo

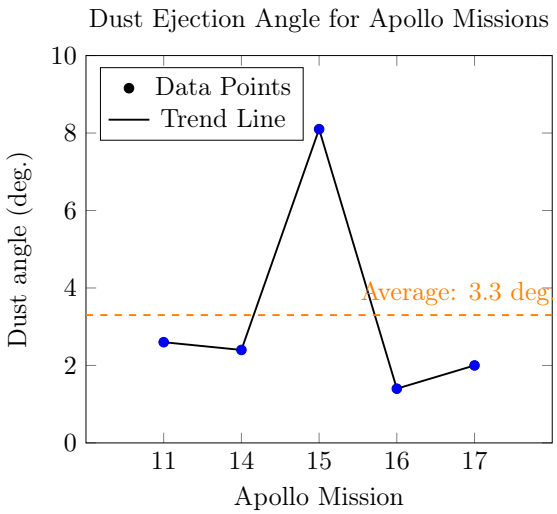


Figure 31: Scatter Plot of Dust Ejection Angles from Apollo Missions with Average Angle. Interpreted from C. Immer et al. (2014).

Appendix E Mission Impact Summary

Hazard type	Hazard amount recorded
Vision	8
Equipment clog	15
Dust coating	25
Slip	4
Inhalation	7
Radiators heating up	5
False sensor readings	3
Seal failures	3
Dust abrasion	4

Table 14: Apollo 17 Lunar Dust Mission Impact Summary. Adapted from J. Gaier (2007).

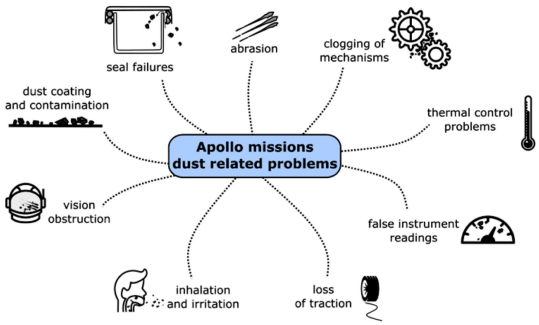


Figure 32: Apollo 17 Lunar Dust Mission Impact Illustration. Provided by Zanon et al. (2023).

Appendix F Matlab Output Nozzle 2 Isentropic Values Full and Quarter Scale

```
>> Nozzle2FullScale
Full Scale Nozzle Contour based on Rocket Plume Interacting with Mars Soil Particulates
Exit Mach number: 7.5265
Area ratio (expansion ratio): 144.1153
Supersonic static inlet pressure: 3079.1191 Pa
```

Figure 33: Nozzle 2 Full Scale

```
>> Nozzle2QuarterScale
0.25x Scale Nozzle Contour based on Rocket Plume Interacting with Mars Soil Particulates.
Exit Mach number: 7.5265
Area ratio (expansion ratio): 144.1153
Supersonic static inlet pressure: 3079.1191 Pa
```

Figure 34: Nozzle 2 Quarter Scale

Appendix G MATLAB Code Employed for MicroJets in 2nd Nozzle

```
% Constants
gamma = 1.4; % Specific heat ratio for Air
R = 287; % Specific gas constant for Air in J/(kg·K)

% Main nozzle conditions
P0_main = 20.2e6; % Total pressure in Pa
T0_main = 3300; % Total temperature in K
P_static_main = 3000; % Supersonic static pressure in Pa

% Simulated data from Initial Martian Descent Thruster Nozzle simulation.
% Velocity profile along the nozzle axis (in m/s)
velocity_profile = linspace(100, 800, 100);
% Pressure profile along the nozzle axis (in Pa)
pressure_profile = linspace(1e5, 8e5, 100);
% Position along the nozzle axis (in meters)
x = linspace(0, 0.5, 100);

% Function to compute static temperature and velocity
function [T_static, V] = compute_isentropic_values(T0, P0, M, gamma, R)
    % Compute static temperature
    T_static = T0 / (1 + (gamma - 1) / 2 * M^2);

    % Compute speed of sound
    a = sqrt(gamma * R * T_static);

    % Compute velocity
    V = M * a;
end

% Function to adjust microjet conditions based on nozzle data
function [P0_microjet, T_static_microjet, V_microjet] = adjust_microjet_conditions(velocity_profile, pressure_profile, x, P0_main, T0_main, gamma, R, station_positions)

    P0_microjet = zeros(1, length(station_positions));
    T_static_microjet = zeros(1, length(station_positions));
    V_microjet = zeros(1, length(station_positions));

    for i = 1:length(station_positions)
        % Find the nearest data point in the nozzle profile
        [~, idx] = min(abs(x - station_positions(i)));

        % Use the velocity and pressure at the station position
        velocity = velocity_profile(idx);
        pressure = pressure_profile(idx);
```

Figure 35: Nozzle 2 Matlab Code for Integration of Microjets

```
        % Set total pressure for microjets
        P0_microjet(i) = 0.1 * P0_main;

        % Set total temperature the same as main nozzle
        T0_microjet = T0_main;

        % Calculate Mach number based on local conditions
        a = sqrt(gamma * R * T0_microjet); % Speed of sound at total temperature
        M_microjet = velocity / a;

        % Calculate static temperature and velocity for microjets
        [T_static_microjet(i), V_microjet(i)] = compute_isentropic_values(T0_microjet, P0_microjet(i), M_microjet, gamma, R);
    end
end

% Define positions of microjet stations outside and above the nozzle exit
station_positions = [0.355, 0.335]; % Positions along the nozzle axis (in mm)

% Adjust microjet conditions based on nozzle data
[P0_microjet, T_static_microjet, V_microjet] = adjust_microjet_conditions(velocity_profile, pressure_profile, x, P0_main, T0_main, gamma, R, station_positions);


fprintf('Microjet Conditions:\n');
for i = 1:length(station_positions)
    fprintf('Station %d (Position: %.3f m):\n', i, station_positions(i));
    fprintf('  Total Pressure: %.2f MPa\n', P0_microjet(i) / 1e6);
    fprintf('  Static Temperature: %.2f K\n', T_static_microjet(i));
    fprintf('  Velocity: %.2f m/s\n', V_microjet(i));
end

% Compute turbulent viscosity ratio
turb_viscosity_ratio = 10;

fprintf('Turbulent Viscosity Ratio: %.2f\n', turb_viscosity_ratio);


function write_microjet_conditions_to_file(station_positions, P0_microjet, T_static_microjet, V_microjet, turb_viscosity_ratio)
    fid = fopen('microjet_conditions.txt', 'w');
    fprintf(fid, 'Microjet Conditions:\n');
    for i = 1:length(station_positions)
        fprintf(fid, 'Station %d (Position: %.3f m):\n', i, station_positions(i));
        fprintf(fid, '  Total Pressure: %.2f MPa\n', P0_microjet(i) / 1e6);
        fprintf(fid, '  Static Temperature: %.2f K\n', T_static_microjet(i));
        fprintf(fid, '  Velocity: %.2f m/s\n', V_microjet(i));
    end
    fprintf(fid, 'Turbulent Viscosity Ratio: %.2f\n', turb_viscosity_ratio);
    fclose(fid);
end

write_microjet_conditions_to_file(station_positions, P0_microjet, T_static_microjet, V_microjet, turb_viscosity_ratio);
```

Figure 36: Nozzle 2 Matlab Code for Integration of Microjets